

Monte Carlo simulation of background and 511-keV point-source sensitivity for a balloon-borne Laue-lens TES telescope

Abstract Focused observations of compact or unresolved Galactic 511 keV emission with a balloon-borne Laue lens require detector backgrounds to be controlled within a sub-keV energy window. We developed an end-to-end Geant4/MEGALib simulation in which the focused signal, prompt atmospheric backgrounds, and activation decays share a common detector-response and event-selection framework for a transition-edge sensor (TES) microcalorimeter focal plane. We first evaluated a reference detector–cryostat configuration. Its day-15 post-selection background rate was $5.31 \times 10^{-2} \text{ s}^{-1}$ and was dominated by delayed activation. Selected-event tracing showed that copper contributed 70.87% of the delayed background, principally from the 50 mK cold plate, TES-support copper panels, and TES copper heat-sink ring. The concentration of selected events in these near-field structures motivated a lateral-chimney configuration, in which the TES was displaced from the cold-stage projection and surrounded by bismuth germanate (BGO) active shielding. This configuration retained 99.75% of the reference post-selection effective area while reducing the day-15 background to $1.07 \times 10^{-2} \text{ s}^{-1}$. Its residual background comprised atmospheric gamma rays (49.8%), delayed activation (33.3%), and non-gamma prompt particles (16.9%). For a 20 d on-source exposure, the top-of-atmosphere 3σ minimum resolvable line flux was $(3.09 \pm 0.28) \times 10^{-5} \text{ ph cm}^{-2} \text{ s}^{-1}$, a factor of 2.23 below that of the reference configuration. The results support two design priorities for low-background 511 keV telescopes: separating the TES from activation-prone near-field materials and their unshielded lines of sight, and increasing the solid angle covered by active anticoincidence.

Keywords Gamma-ray instrumentation, 511 keV annihilation line, transition-edge sensors, Laue lens, balloon-borne telescope, activation background

1 Introduction

Galactic 511 keV annihilation radiation is a key observational probe of the production, propagation, and annihilation environments of Galactic positrons. Its spectrum contains a two-photon line near 511 keV and an ortho-positronium continuum at lower energies. Positrons annihilate directly with electrons or form para-positronium, producing two photons near 511 keV, whereas ortho-positronium decays into a three-photon continuum. The line intensity, profile, and positronium continuum therefore constrain the medium in which positrons slow down and annihilate [1, 2]. Under an approximate steady-state assumption, the total Galactic 511 keV flux measured by INTEGRAL/SPI corresponds to an annihilation rate of a few $\times 10^{43} \text{ e}^+ \text{ s}^{-1}$ [1, 3]. Proposed positron sources include β^+ -unstable nuclei synthesized in stellar explosions, compact objects capable of producing electron–positron pairs, and more speculative particle-physics scenarios in which annihilating or decaying light dark matter injects low-energy positrons. Existing observations do not uniquely determine the dominant contribution [1, 4, 5].

Wide-field observations have established the diffuse morphology of the line. INTEGRAL/SPI finds a bright central bulge and a weaker disk, and a recent 20-year analysis recovers central, broad-bulge, and disk components while the evidence for additional correlated structures remains marginal [4, 6, 7]. The 2016 COSI balloon flight independently detected the bulge and found evidence that the emission is more extended than a single point source [3, 8]. Spectroscopy also constrains the annihilation medium and kinematics: the SPI spectrum can be decomposed into a narrow component with about 1.3 keV FWHM and a broad component with about 5.4 keV FWHM, and spatial variations in centroid energy and width have been sought across the inner Galaxy [2, 9]. The annihilation map, however, records where positrons slow down and annihilate, not necessarily where they were created. Propagation through a magnetized, multiphase interstellar medium means that diffuse morphology and spectroscopy alone cannot recover the dominant source class or the complete pre-annihilation transport

history [1].

Wide-field Compton and coded-mask instruments are well suited to the total bulge and disk emission. A complementary pointed instrument is needed to test whether the central region also contains an unresolved central enhancement, compact sources, or measurable line-kinematic structure. A detection of a point-like or narrowly concentrated feature would show that part of the annihilation is spatially localized; a non-detection would constrain a compact component without excluding source populations whose positrons escape and annihilate diffusely. An INTEGRAL/IBIS compact-source search found no Galactic 511 keV point sources and set a 2σ upper limit of about $1.6 \times 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$ for a ~ 10 Ms Galactic-centre exposure [10]. This limit provides a benchmark for a narrow-field pointing instrument.

Laue optics use transmission-mode Bragg diffraction to focus photons arriving from a known direction onto a compact focal plane [11, 12]. When detector-related background dominates, concentrating photons collected over a large aperture onto a small sensitive detector area improves point-source sensitivity [13]. The focal-plane detector must combine a narrow energy response with sufficient stopping efficiency at 511 keV. Transition-edge-sensor (TES) microcalorimeters measure the small temperature rise produced by deposited energy through the steep resistive transition of a superconductor and can achieve sub-keV energy resolution near 511 keV [14, 15]. An array covers the focal spot, while thick absorbers and a multilayer layout increase the full-energy-peak efficiency. The Laue lens concentrates source photons onto a compact focal plane, and the TES response supports a narrow line window and detailed line-profile measurements. The resulting narrow-field focusing spectrometer trades sky coverage for a concentrated source beam and high-resolution spectroscopy. It complements previous wide-field measurements of the Galactic bulge and disk and supports dedicated searches for compact sources.

Energy resolution and focusing performance alone do not determine point-source sensitivity [14, 16]. Prompt energy deposits from the radiation environment and delayed emission from activated detector and cryostat materials also set the background in the science window [17, 18].

This work evaluates the high-altitude performance of a balloon-borne Laue-lens TES-array telescope for detecting 511 keV point sources toward the Galactic center. We classify the science-window background by physical component and incident particle, trace it to the relevant materials, parent nuclides, and production regions, and use this provenance to identify the dominant production mechanisms and guide optimization of the detection system.

1.1 Science and instrument requirements

The instrument simulated here is designed to search for compact 511 keV sources and an unresolved central-excess component in the Galactic-center region. INTEGRAL/SPI measurements show that the 511 keV sky is dominated by diffuse bulge and disk emission [4, 7]. De Cesare's dedicated search for compact sources with INTEGRAL/IBIS found no Galactic 511 keV point sources and reported a 2σ upper limit for a 10 Ms exposure toward the Galactic center [10]. For the mission fold, we rescale this limit linearly from 2σ to 3σ under the Gaussian approximation and set the top-of-atmosphere reference flux to $F_{\text{ref}} = 2.4 \times 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$. At each trajectory node, the mission fold calculates atmospheric transmission from the residual column depth at the assumed balloon altitude and a fixed 27° reference elevation. The representative reference state is the day-15 trajectory node at 38.75 km; the adopted residual column at this node gives $T_{\text{ref}} = 0.514$. The corresponding telescope-incident reference flux is $F'_{\text{ref}} = F_{\text{ref}} T_{\text{ref}} = 1.233 \times 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$. At other trajectory nodes, the incident flux is $F_{\text{ref}} T_{\text{atm},511}(t)$, and minimum resolvable fluxes are reported before atmospheric attenuation. Compared with the wide-field spectroscopic and coded-mask searches conducted with INTEGRAL, the payload modeled here uses Laue focusing for its point-source response and a high-energy-resolution, multilayer thick-absorber TES microcalorimeter array at the focal plane [14, 19].

Sections 2–4 define the mass models and simulation framework and compare their balloon background and point-source sensitivity. Section 5 screens the model-B response in LEO, lunar-surface, and Sun–Earth L2 environments; Sections 6 and 7 summarize the design implications and next steps.

2 Detector and cryostat mass model

This study compares two detector–cryostat configurations: reference model A and lateral-chimney model B. Both contain the same six-layer TES core and a compact dilution refrigerator (DR) with a mixing chamber, staged cold plates, thermal and magnetic shields, copper supports, active scintillators, and cryogenic service

structures [14, 20]. They differ in the placement of the focal plane and in the physical entrance, local tungsten structure, and active shield arrangement. Figure 1 compares the two geometries with the same camera and physical scale.

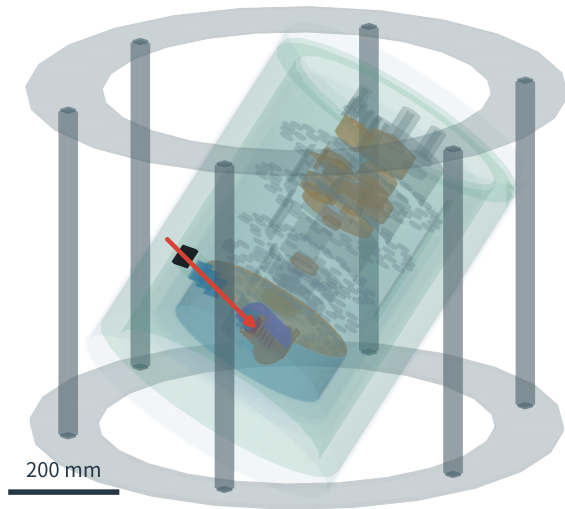
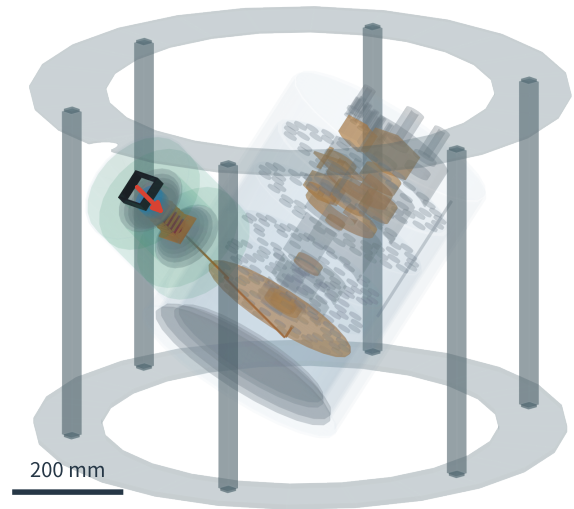
The TES detector concept uses an absorber-coupled AlMn-film structure developed in our laboratory. The transition temperature of AlMn can be tuned through its composition and annealing process; the material is suitable for array fabrication and has relatively low magnetic-field sensitivity [22, 23]. In the present conceptual design, the TES thermometer is an approximately 200 nm-thick AlMn film with an Mn doping concentration of 2000 ppm on an atomic basis, whereas the γ -ray absorber is a millimetre-scale metal body. The absorber consists of $1.5 \times 1.5 \times 3.0 \text{ mm}^3$ Ta pixels; the high atomic number of Ta increases the interaction efficiency for 511 keV photons. At 511 keV, a single 3.0 mm Ta layer has an estimated interaction efficiency of about 49%, and the corresponding six-layer interaction probability is about 98%. In the current geometry, adjacent Ta array layers have a center-to-center spacing of 12 mm; this spacing was chosen with a subsequent veto based on Compton-kinematic track reconstruction in mind. Ta was also selected for its low superconducting specific heat in the cryogenic regime ($5.99 \times 10^{-7} \text{ J kg}^{-1} \text{ K}^{-1}$ at 100 mK [21]). Ta also has a relatively high photoelectric-to-Compton interaction ratio. From the NIST XCOM cross sections near 511 keV, this ratio is approximately 0.731 [24]. In the detector-response model, we use our laboratory experience with a previously developed X-ray TES [22] to adopt $\Delta E_{\text{FWHM}} \simeq 420 \text{ eV}$ at 511 keV as the estimated energy resolution and detector-response post-processing input.

Because the TES film and Ta absorber differ greatly in dimensions and mass, and because the TES film itself contributes negligibly to interactions at 511 keV, the transport geometry uses the Ta absorber as the principal sensitive volume of each TES pixel. The explicitly modelled sensitive volume comprises six layers, each containing 376 Ta absorber pixels, for a total of 2256 pixels. Each pixel is 1.5 mm long and wide and 3 mm thick. A Si substrate proxy is placed below each Ta pixel; because detailed thermal conduction is not modelled, Si represents the $\text{SiO}_2\text{--SiN}_x\text{--Si}$ substrate. The TES film, wiring, and readout details are not constructed layer by layer as explicit interaction volumes, but enter the model through the assumed energy response and service-mass proxies.

The TES array in mass model A is mounted on copper supports immediately below the staged cold plates. Along the optical path are successive aluminium thermal-shield windows, a beryllium vacuum window, and a tungsten multihole collimator aligned with the TES pixels. Boolean apertures through the surrounding shells define the side-entry beam path.

Source tracing in model A identified the 50 mK mixing-chamber copper plate, TES copper supports, and copper heat sink as important 511-keV background sources. Model B therefore moves the same six-layer TES core beyond the cold-plate projection and surrounds it with local BGO, aiming to reduce cold-end coupling and active-shield mass.

Both designs use the same MEGAlib/Geant4 framework, source fields, TES response, timing, and selection algorithms; active channels and physical apertures are geometry-specific.

a Mass model A: TES below cold stages**b Mass model B: TES in lateral chimney**

■ active BGO ■ outer support frame ■ Bi shield (A) ■ W collimator / frame
■ Al / cryostat envelopes ■ cold Cu structures ■ TES Ta absorbers

Figure 1: Matched detector–cryostat views of (a) mass model A and (b) mass model B, rendered with the same viewing angle and scale. In model A, the six-layer TES array lies below the staged cold structures and behind a multihole W collimator. Model B translates the same TES core into a lateral Al side chimney, with local active BGO on the chimney side, front optical annulus, and rear cold-port annulus. Red arrows show the nominal focused 511 keV paths; selected outer envelopes are translucent or omitted for clarity.

3 Simulation framework

Particle transport was performed with Cosima in MEGAlib 4.02.00 (Geant4 10.02.p03), using the LivermorePol electromagnetic and qqsp-bic-hp hadronic physics lists [25, 26, 30]. Cosima defines sources and detector geometry, transports particles through the mass model, and records interactions for detector-response and event-selection analysis. A uniform Geant4 production range cut of $5\ \mu\text{m}$ was used. Figure 2 summarizes the end-to-end calculation in three physical branches: focused signal, prompt atmospheric background, and delayed activation background. The Laue optics and the detector–cryostat system were represented by separate mass models. The reference point source was first transported through the Laue-lens model; the positions and directions of photons reaching the focal plane were written to a MEGAlib EventList and used to initialize detector-side Monte Carlo transport. The focused signal and all backgrounds were transported through each detector–cryostat mass model.

Prompt transport and activation-production transport were run in parallel from the same eight-family EX-PACS/PARMA atmospheric field, comprising γ , e^- , e^+ , μ^- , μ^+ , n , p , and α . In the delayed-activation branch, BUILDUP transport recorded nuclide yields, material volumes, and production positions. These records were used to calculate the activity accumulated during the flight and to construct the radionuclide inventory at each reference epoch; delayed decays were then sampled at the recorded production positions and transported through the detector–cryostat model.

The focused signal, eight prompt families, and delayed activation use the same detector response within each model. The science-window cut is followed by the geometry-specific active veto and common Compton-topology test. Selected backgrounds are decomposed by incident particle, parent nuclide, material, and production volume before the mission fold. The subsections below define the source models, transport normalization, event selection, and mission-time calculation.

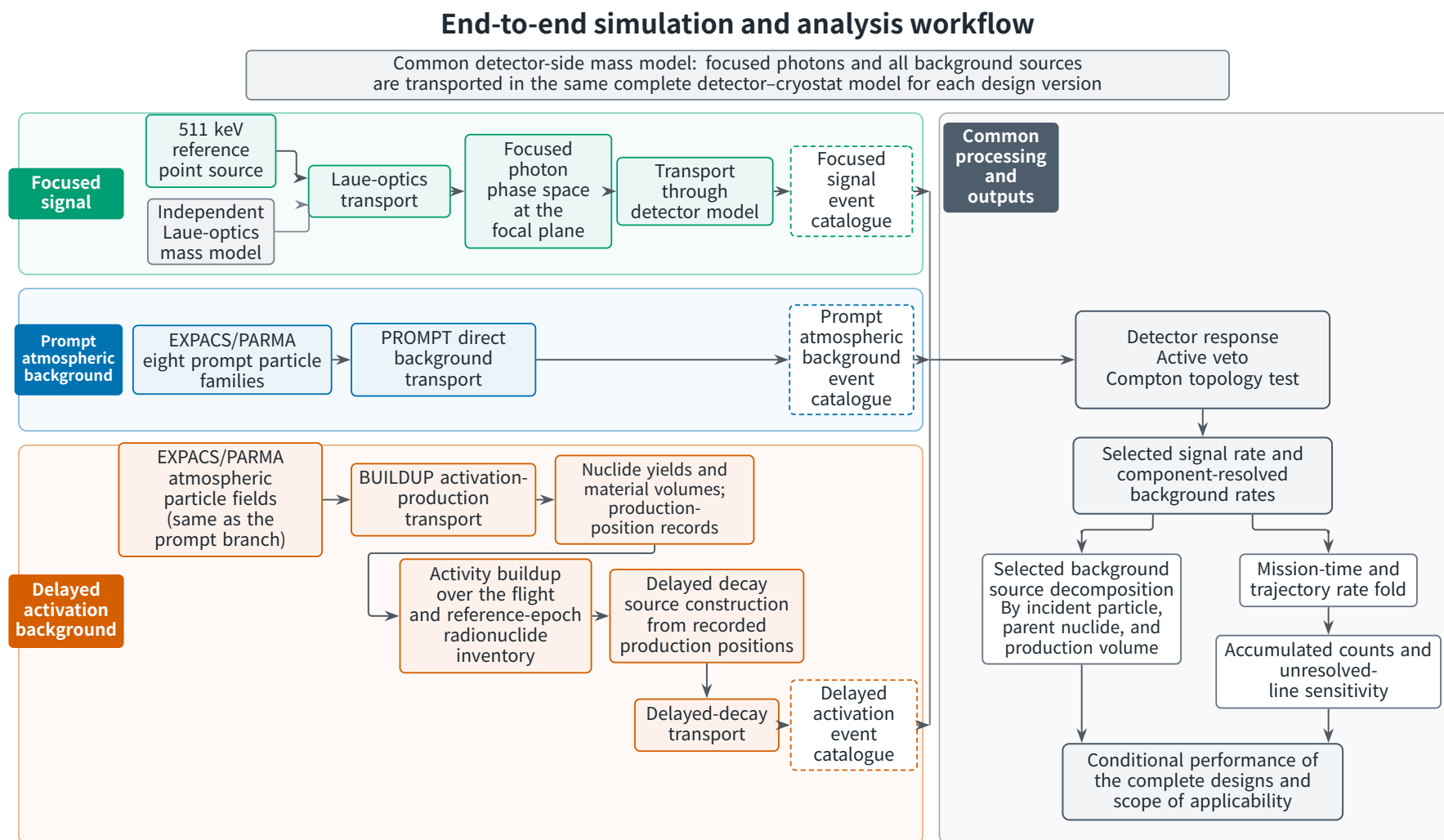


Figure 2: End-to-end simulation and analysis workflow. The focused signal, prompt atmospheric background, and delayed activation form three physical branches. The prompt branch comprises eight independently normalized particle families (γ , e^- , e^+ , μ^- , μ^+ , n , p , α). All streams undergo the common detector response, active veto, and Compton-topology test within each mass model. Selected-event provenance motivates model B; both designs are then evaluated on the common mission timeline.

3.1 Laue optics and detector-coupled signal transport

The simulated Laue lens uses mosaic Ge(111) crystals [11, 12]. The optical design has a 10 m focal length, a single-ring reference response at 511 keV, and an effective area of 20.1 cm² at 511 keV, calculated as

$$A_{\text{eff}}(E) = A_{\text{inc}} \frac{N_{\text{fp}}(E)}{N_{\text{inc}}(E)} \quad (1)$$

where A_{inc} is the Monte Carlo incident sampling area, $N_{\text{inc}}(E)$ is the number of incident photons, and $N_{\text{fp}}(E)$ is the number of photons satisfying the focusing/focal-plane interception condition. The mosaic-crystal implementation provides the specified angular acceptance and is used consistently throughout the optical transport.

Diffraction is implemented as a discrete application-layer process in Geant4 rather than as an efficiency applied after transport [25, 26]. At angular offset $|\Delta\theta|$, the tabulated diffraction probability R is first normalized to the unabsorbed fraction, $R_{\text{eff}} = R/(1 - A)$, where A is the photoelectric-absorption probability. It is then converted to an equivalent mean free path λ_{diff} through $1 - \exp(-\ell/\lambda_{\text{diff}}) = R_{\text{eff}}$, where ℓ is the photon path length through the crystal step; hence $\lambda_{\text{diff}} = -\ell/\ln(1 - R_{\text{eff}})$. Geant4 samples this process together with photoelectric absorption and Compton scattering, and the shortest sampled interaction length determines which process occurs first. A photon not selected for diffraction therefore continues under the standard electromagnetic physics and may be absorbed, scattered, or transmitted.

In this work, the diffraction probability is determined from a precomputed XOP/CRYSTAL rocking curve. The angular dependences of the Ge(111) reflectivity, transmissivity, and absorption are generated with the CRYSTAL/diff_pat module of the XOP toolkit (called through xoppylib using crystallographic data from DABAX) [27] and stored as a transport lookup table during preprocessing. At each transport step inside a crystal, the angular deviation from the Bragg condition is evaluated and the corresponding diffraction probability is obtained by interpolation. The 30 arcsec FWHM mosaic spread is implemented by applying a Gaussian angular perturbation to the ideal Bragg-plane normal. When diffraction occurs, the outgoing direction is determined from the perturbed plane normal, while the photon energy is conserved. This treatment follows existing Geant4 implementations of crystal Bragg reflection [28, 29] and is applied to the energy range considered here.

Here the reference signal is a monochromatic 511 keV point source. Its sole reference normalization is $F_{\text{ref}} = 2.4 \times 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$ above the atmosphere (Section 1.1), motivated by the dedicated INTEGRAL/IBIS compact-source upper limit [10] rather than by any single observed object. It is constructed as a MEGALib far-field source [30], with the detailed construction given in Section 3.2.3. The signal is transported through the optics branch and injected into the detector mass model as a parallel-beam source; prompt cosmic-ray background and activation decays are transported directly in the detector/cryostat mass model. The primary background budget is therefore defined for the detector/cryostat branch, while the optics branch supplies the transported focal-plane signal phase space.

3.2 Source construction

The reference point-source EventList, the prompt atmospheric field, and the delayed-decay source are constructed independently. The eight prompt particle families retain separate physical-rate normalizations; activation production and delayed decay remain separate transport stages. This organization preserves the normalization and provenance needed to compare all selected background streams under the common detector response and event selection.

3.2.1 Prompt atmospheric source

The prompt atmospheric field is derived from EXPACS/PARMA. PARMA3.0 is an analytical model parameterized from PHITS atmospheric-shower calculations and benchmarked against terrestrial cosmic-ray measurements, and EXPACS is its public tabulation and software interface [31, 32]. PHITS supplies the underlying particle-transport physics: primary cosmic rays interact with air nuclei, the secondary hadrons, leptons, photons, and neutrons are transported through the atmospheric column, and the resulting fluences are parameterized by altitude, geomagnetic cutoff, solar modulation, energy, and direction. In this work EXPACS/PARMA provides the differential incident flux as a function of particle species, kinetic energy, zenith angle, geographic

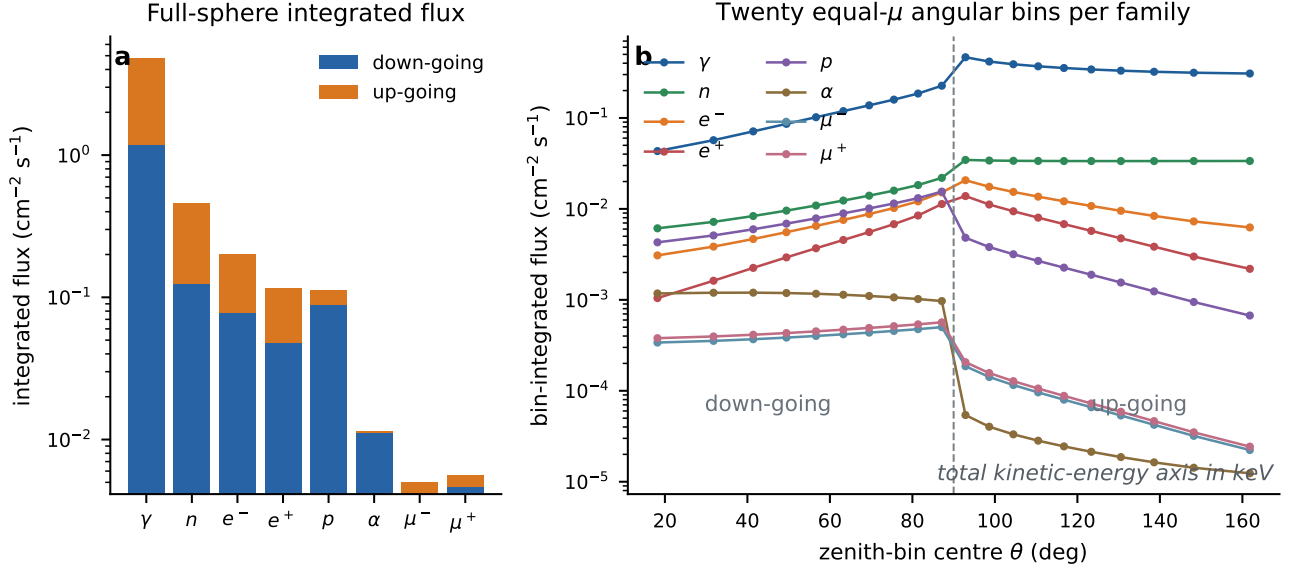


Figure 3: EXPACS/PARMA atmospheric particle field used for prompt transport and activation production. (a) Fluxes of the eight particle families integrated over energy and solid angle in the down-going and up-going (albedo) hemispheres. (b) Integrated fluxes in the 20 equal- μ zenith bins ($\mu = \cos \theta$) for each particle family. The environment is latitude 34° , longitude 100° , altitude 38 km, $R_c = 11.6$ GV, and $W = 118.3$.

position, altitude, cutoff rigidity, and solar activity, and is used only to construct the external radiation fields; detector response, vetoing, topology selection, and mission-time counting are applied downstream. The transport templates use latitude 34° , longitude 100° , altitude 38 km, vertical cutoff rigidity $R_c = 11.6$ GV, and the 2025-08-31 solar condition corresponding to $W = 118.3$.

The prompt source includes photons, neutrons, electrons, positrons, protons, alpha particles, and negative and positive muons. For each particle family the EXPACS/PARMA differential spectrum is converted into a MEGALib far-field area source covering the full zenith range $\theta = 0^\circ$ – 180° and full azimuth range $\phi = 0^\circ$ – 360° . The zenith dependence is represented by 20 equal- μ bins, where $\mu = \cos \theta$: the first ten are down-going and the last ten up-going or albedo-like. Both hemispheres are retained in the prompt transport and in the activation-production transport, so upward albedo particles are not discarded at source construction. Figure 3 shows the resulting full-sphere flux field.

The full-sphere integrated fluxes in the adopted atmospheric tables are 4.800, 0.4622, 0.1990, 0.1170, 0.1123, 0.01149, 0.004969, and 0.005570 $\text{cm}^{-2} \text{s}^{-1}$ for γ , n , e^- , e^+ , p , α , μ^- , and μ^+ , respectively.

If $I_j(E, \mu)$ denotes the EXPACS/PARMA differential intensity for family j , the flux carried by equal- μ bin m is

$$\Phi_{j,m} = 2\pi \int_{\mu_m}^{\mu_{m+1}} d\mu \int I_j(E, \mu) dE, \quad \Delta\mu = 0.1, \quad (2)$$

so each bin subtends $\Delta\Omega = 2\pi\Delta\mu = 0.6283$ sr.

All eight particle families use a far-field area source of radius $R = 60$ cm. For each sampled direction, starting positions are drawn uniformly from the projected disk perpendicular to that direction, and momenta point into the mass model; the geometric rate factor is the projected area πR^2 . Together, the 20 bins cover 4π incident solid angle. The same atmospheric-field definition is used separately for prompt transport and activation-production calculations.

Each accepted production job for particle family j contains the complete 20-bin angular mixture; the angular bins are not treated as independent exposures. The family source rate and the merged equivalent exposure are therefore

$$\dot{N}_j^{\text{src}} = \pi R^2 \sum_{m=1}^{20} \Phi_{j,m}, \quad T_{g,j}^{\text{eq}} = \sum_{q \in \mathcal{Q}_{g,j}} \text{TT}_{g,j,q} \simeq \frac{\sum_q N_{g,j,q}^{\text{sim}}}{\dot{N}_j^{\text{src}}}, \quad w_{g,j}^{\text{pr}} = \frac{1}{T_{g,j}^{\text{eq}}}, \quad (3)$$

where q indexes independent mergeable jobs and $\text{TT}_{g,j,q}$ is the validated physical runtime recorded in the cor-

responding receipt. Increasing the number of transported primaries extends T^{eq} and reduces Monte Carlo uncertainty without changing the physical source intensity.

Equation (3) applies directly to the seven non- γ families. Prompt atmospheric photons were transported from the corrected-keV 20-bin broadband total spectrum and each detector-positive record was reweighted once to the de-lined continuum as $w_e = (T_{g,\gamma}^{\text{eq}})^{-1} J_{\text{cont}}(m_e, E_e) / J_{\text{total}}(m_e, E_e)$. Here $m_e = \text{clip}[\lfloor 10(1 + d_{z,e}) \rfloor, 0, 19]$, and both spectra were linearly interpolated on the same energy grid. Validation confirmed source-flux closure and complete incident-record matching for both models.

3.2.2 Delayed activation source

The delayed source is built from a separate activation-production transport. It is not counted as immediate detector background; instead, it records the isotope, material volume, incident family, and production position of each radioactive product. Ground-state half-lives are checked against NUBASE2020 [33]. For isotope k ,

$$A_k(t_{\text{flight}}) = P_k [1 - \exp(-\lambda_k t_{\text{flight}})], \quad \lambda_k = \frac{\ln 2}{t_{1/2,k}}, \quad (4)$$

where P_k is the production rate and the reference state is $t_{\text{flight}} = 15$ d.

In the standard output configuration used here, MEGALib/Cosima's compressed `sim.gz` event files do not retain the complete radioactive-product production information required to reconstruct the spatial distribution of the delayed source. Because Cosima uses Geant4 as its transport engine, we added a recording hook to `SteppingAction` to store the production coordinate, material volume, nuclide, and excitation state when each radioactive product was created. These fields were then used to match the day-15 inventory to the production records and to organize the delayed source by incident family, material category, production volume, and coordinate.

Retaining the production positions is essential because the ability of a delayed particle to escape the surrounding material, trigger the geometry-specific active anticoincidence, or pass the Compton-topology selection depends on where it was produced. Balancing computational cost against statistical precision, we constructed 10^4 activation-position point sources for each incident-particle family. Sampling was weighted by the day-15 activity of each radionuclide in its production volume, and the decay position was drawn with equal probability from the matched simulated production coordinates. The activation-source positions in the two mass models, together with the parent nuclides contributing to the TES 511 keV response window, are shown in Fig. 4.

For each incident-particle family, each mass model transports $N_{g,j}^{\text{dec}} = 10^6$ delayed decays. At the day-15 reference epoch, the constant-activity equivalent exposure and per-event rate weight of that family are

$$T_{g,j}^{\text{eq,del}}(t_{\text{ref}}) = \frac{N_{g,j}^{\text{dec}}}{A_{g,j}(t_{\text{ref}})}, \quad w_{g,j}^{\text{del}}(t_{\text{ref}}) = \frac{A_{g,j}(t_{\text{ref}})}{N_{g,j}^{\text{dec}}} = \frac{1}{T_{g,j}^{\text{eq,del}}(t_{\text{ref}})}, \quad t_{\text{ref}} = 15 \text{ d}. \quad (5)$$

Here $T^{\text{eq,del}}$ is the Monte Carlo equivalent sampling time for the day-15 activity mixture, rather than the elapsed flight time or the BUILDUP irradiation time; the mission fold updates the day-15 reference weights at each time node using the parent-nuclide activity curves.

Table 1 summarizes the prompt and delayed transport statistics and their rate normalizations.

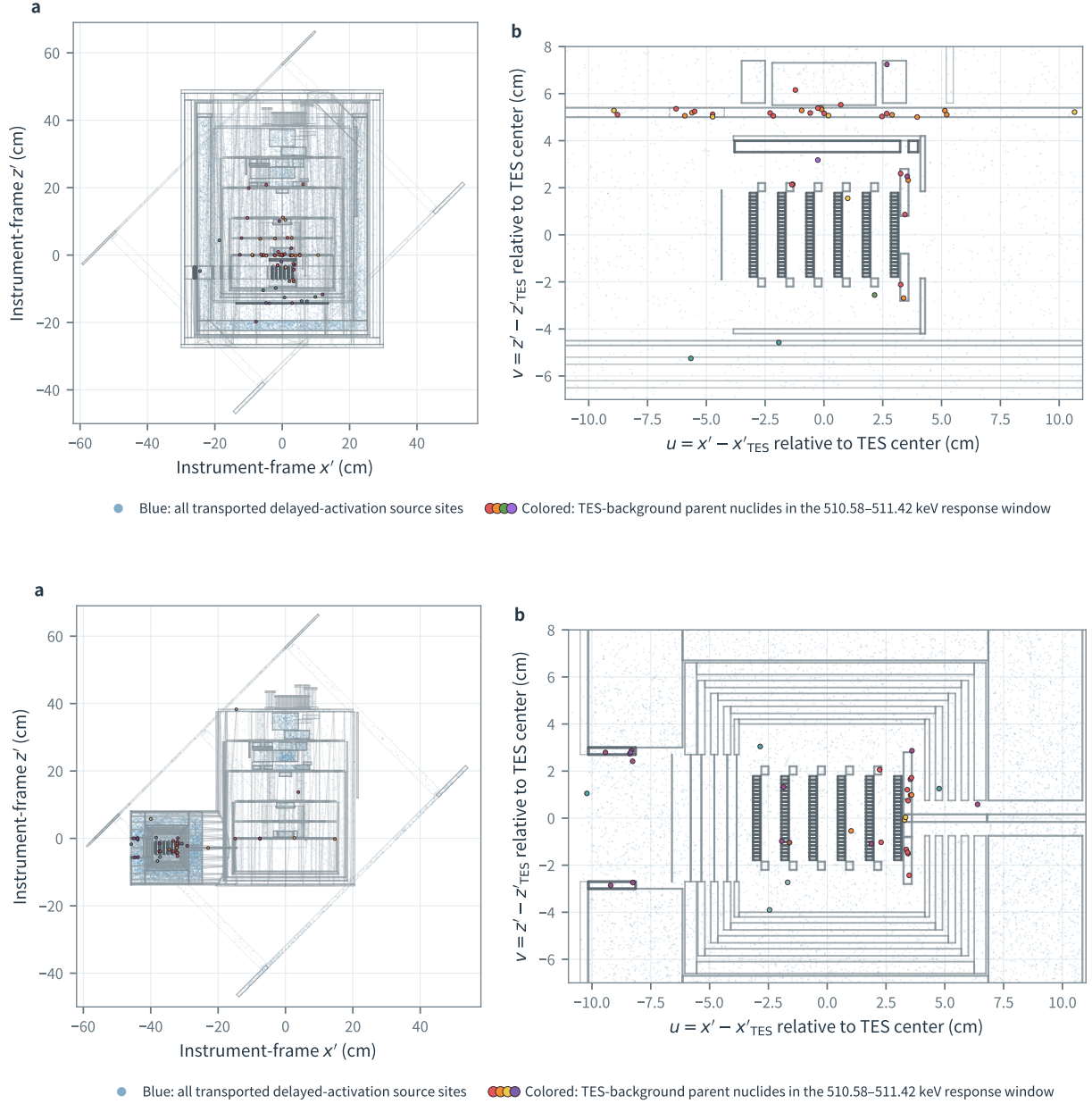


Figure 4: Delayed activation-production positions in mass model A (top) and mass model B (bottom). In each row, the left panel shows the full mass model and the right panel enlarges the TES region. Blue points denote all transported delayed activation-source positions; coloured points denote activation parent nuclides that produce background in the TES 510.58–511.42 keV response window.

Table 1: Particle-family transport statistics and rate normalization for the two mass models. N_{tr} and N_{dec} are transported prompt primaries and delayed decays, respectively; T_{eq} is the summed physical exposure of accepted independent production batches; and $N_{\text{det+}}$ counts templates with a positive raw deposit in the TES or an active shield. R_{cat} is the detector-positive occupancy rate actually supplied to common-time Poisson sampling. It is evaluated before detector-response smearing, the pixel threshold, science window, active veto, and Compton selection. In panel (b), $T_{\text{eq,del}} = N_{\text{dec}}/A_{g,j}(15 \text{ d})$, as defined in Equation (5).

(a) Prompt atmospheric-particle transport					
Model	Family	N_{tr}	$T_{\text{eq}} \text{ (s)}$	$N_{\text{det+}}$	$R_{\text{cat}} \text{ (s}^{-1}\text{)}$
A	γ	20,315,706	374.2	6,372,612	1.636×10^4
A	n	1,164,960	223.1	403,599	1.809×10^3
A	e^-	675,057	299.8	287,509	959.1
A	e^+	394,119	298.9	182,318	610.0
A	p	380,829	299.8	172,734	576.3
A	α	39,050	299.0	18,657	62.41
A	μ^-	16,607	298.2	7,529	25.25
A	μ^+	19,371	308.7	8,629	27.96
B	γ	112,421,262	2071	4,189,660	1.940×10^3
B	n	8,363,786	1599	325,382	203.4
B	e^-	3,602,914	1599	136,738	85.52
B	e^+	2,119,548	1601	139,963	87.42
B	p	2,036,710	1603	123,624	77.13
B	α	209,074	1608	18,285	11.37
B	μ^-	91,020	1608	4,605	2.864
B	μ^+	102,306	1613	5,268	3.266
(b) Delayed-decay transport at day 15					
Model	Family	N_{dec}	$T_{\text{eq,del}} \text{ (s)}$	$N_{\text{det+}}$	$R_{\text{cat},15} \text{ (s}^{-1}\text{)}$
A	γ	1.0×10^6	1.517×10^5	884,678	5.832
A	n	1.0×10^6	3030	893,664	294.9
A	e^-	1.0×10^6	1.497×10^6	853,592	0.5701
A	e^+	1.0×10^6	3.386×10^5	904,775	2.672
A	p	1.0×10^6	1381	916,824	664.0
A	α	1.0×10^6	3478	918,773	264.2
A	μ^-	1.0×10^6	4.034×10^6	871,246	0.2160
A	μ^+	1.0×10^6	7.934×10^{14}	1,000,000	1.260×10^{-9}
B	γ	1.0×10^6	4.921×10^5	396,106	0.8049
B	n	1.0×10^6	1.459×10^4	554,831	38.02
B	e^-	1.0×10^6	3.755×10^6	433,022	0.1153
B	e^+	1.0×10^6	1.481×10^6	531,844	0.3592
B	p	1.0×10^6	9476	694,056	73.24
B	α	1.0×10^6	2.917×10^4	711,199	24.38
B	μ^-	1.0×10^6	1.462×10^7	436,535	0.02986
B	μ^+	1.0×10^6	2.521×10^8	1,867	7.405×10^{-6}

3.2.3 Reference point-source construction

The focused-signal branch uses an on-axis monochromatic 511 keV far-field point source, represented by a parallel beam at the lens entrance. The flux convention and atmospheric attenuation of the reference source are defined in Section 1.1; this subsection specifies only the focal-plane photon phase space and the detector effective-area calculation.

Photons are sampled uniformly over the lens aperture A_{inc} defined by Equation (1). The 30 arcsec mosaic spread, XOP/CRYSTAL rocking curve, and 10 m geometry set the focal-plane response. Three samples of 1.5×10^5 photons yield 37,256 focal-plane crossings, of which $N_{\text{fp}} = 37,194$ fall within the $r = 1.898$ cm beryllium entrance. At 10 m, $r_{90} = 1.03$ cm and $r_{99} = 1.25$ cm correspond to 3.54 and 4.30 arcmin. The accepted photon phase space forms the common EventList for both mass models.

This EventList represents the optical effective area $A_{\text{fp}} \equiv A_{\text{eff}}(511 \text{ keV}) = 20.08 \text{ cm}^2$. Each equally weighted focal-plane photon therefore carries the area weight

$$a_{\text{ray}} = \frac{A_{\text{fp}}}{N_{\text{fp}}}. \quad (6)$$

For mass model g , let $N_{\text{sel},g}$ be the number of focal-plane photons retained after detector transport, response convolution, science-window selection, the geometry-specific isolated-event active-anticoincidence criterion,

and the Compton topology/aperture selection. Both models use the same Compton reconstruction algorithm, while aperture compatibility is tested against the physical entrance of the corresponding model. The final on-axis isolated-event effective area is

$$A_{\text{eff},g} = \sum_{i \in \text{sel},g} a_{\text{ray}} = A_{\text{fp}} \frac{N_{\text{sel},g}}{N_{\text{fp}}}. \quad (7)$$

3.3 Detector response, event selection, and veto definitions

The common-time calculation maps the prompt atmospheric and delayed-activation background catalogues onto one explicit time axis before the active-veto and Compton selections are applied. The eight prompt particle families retain independent physical-rate normalizations. Focused-signal events are inserted as conditional probes of accidental-coincidence loss rather than added to the background budget.

3.3.1 Time axis and rate normalization

In a stationary radiation environment, mutually independent particle events with a constant mean arrival rate have Poisson-distributed counts in a fixed time interval. Conditional on the event count, their arrival times are uniformly distributed within that interval. This provides a natural temporal representation for rate-normalized Monte Carlo catalogues: the catalogue supplies the energy deposits, positions, and topologies, while the Poisson process determines how often and when those events occur on the physical time axis.

The radiation environment evolves along the balloon trajectory, but much more slowly than the $1 \mu\text{s}$ coincidence time scale adopted in this study. Within each trajectory node, the event streams can therefore be approximated as independent Poisson processes with constant arrival rates. The focused signal, prompt atmospheric background, and delayed-activation background are generated by separate transport calculations and do not share a physical time axis, whereas the geometry-specific active anticoincidence and Compton-topology selections depend on all energy deposits within the same time window. Evaluating accidental coincidences therefore requires these catalogues to be mapped onto a common time axis.

The three transported streams acquire per-record rate weights r_{km} during the source-normalization stage of Section 3.2, where $k \in \{\text{signal}, \text{prompt}, \text{delayed}\}$ labels the catalogue and m labels a transported record. The total rate of catalogue k is $R_k = \sum_m r_{km}$. For a reference interval of duration T , the number of instances drawn from catalogue k is

$$N_k \sim \text{Pois}(R_k T). \quad (8)$$

Records are sampled with replacement with probabilities r_{km}/R_k and are assigned independent uniform times in $[0, T]$. The resulting sequence preserves both the physical rate and the energy, direction, and detector-response distribution of the event stream.

The three draws are merged and ordered by arrival time. Events separated by no more than $\tau = 1 \mu\text{s}$ are linked transitively into one coincidence candidate group c . All energy deposits in the group are then subjected jointly to the science-window, geometry-specific active-anticoincidence, and Compton-topology selections. Figure 5 summarizes this construction.

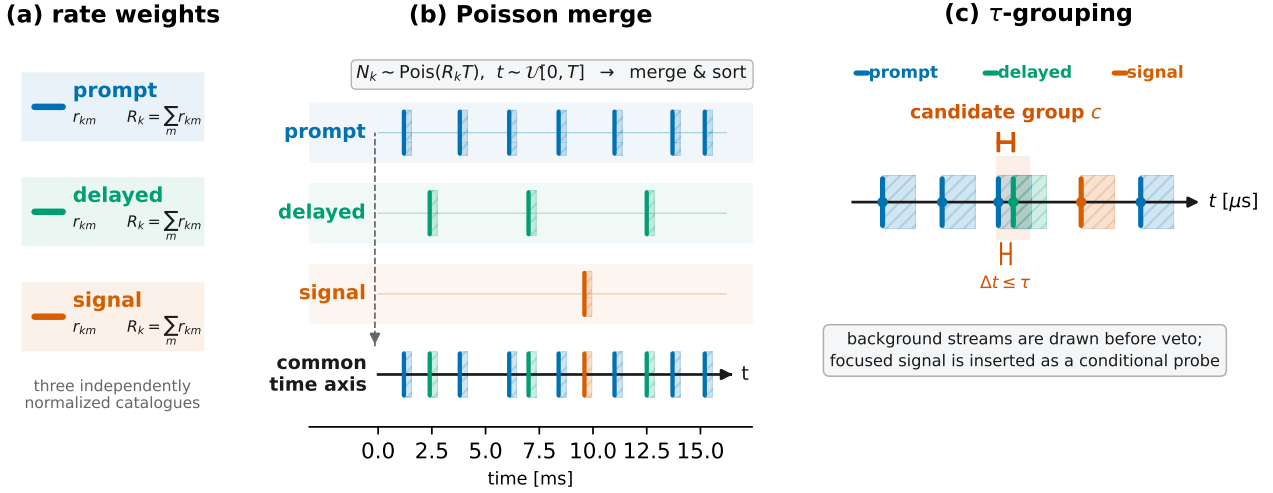


Figure 5: Schematic construction of the common-time coincidence axis. (a) Each transported stream carries per-record rate weights and a total rate. (b) Independent Poisson draws are merged and time-ordered on a common axis. (c) Events separated by no more than $\tau = 1 \mu\text{s}$ form one coincidence candidate group. Colours identify catalogue membership; event positions and densities are schematic. Prompt and delayed streams define the background axis; focused signal is inserted as a conditional probe.

This calculation is performed at five reference times on days 0, 5, 10, 15, and 20. At each reference time, the physical rates of the eight prompt particle families and delayed activation set the Poisson draws used to construct a common background time axis for that reference environment. Events separated by no more than $1 \mu\text{s}$ form one coincidence group and jointly undergo the science-window, geometry-specific active-anticoincidence, and Compton-topology selections. Focused-signal events are also inserted individually into the corresponding background axis to quantify signal loss from accidental coincidences.

Instrument performance is then evaluated along the long-duration balloon trajectory, with the 20 d flight divided into 6 h intervals. Particle fluxes, atmospheric transmission of the signal, and the activation inventory evolve with flight time and position, thereby changing both the event density on the common time axis and the accidental-coincidence probability. The background corrections and signal-survival fractions obtained at the five reference times are applied along the full trajectory. Section 3.4 specifies the trajectory model, temporal interpolation, and cumulative-count calculation.

3.3.2 Detector response, science window, and weighted rates

Before the science-window and anticoincidence decisions, TES energy deposits in each candidate group are combined by pixel. Multiple deposits in the same pixel are summed to give its true deposited energy, and their energy-weighted position represents the interaction location. The expected single-pixel energy resolution is $\Delta E_{\text{FWHM}} = 420 \text{ eV}$; TES measurement fluctuations are approximated by a Gaussian response with this FWHM. Pixel hits with measured energies below 0.300 keV are removed. Mass models A and B use the same response and threshold settings.

The measured energies of all retained TES pixel hits are summed to give the total measured TES energy of the event, which is used for the 511 keV science-window selection. The energy, position, layer, and multiplicity of each pixel hit remain available for the subsequent Compton-topology selection. BGO deposits are accumulated separately, are not included in the TES total, and enter the active anticoincidence decision in the next subsection.

The common science window has a fixed one-FWHM half-width: $510.58 \leq E_{\text{TES}} < 511.42 \text{ keV}$ for the focused signal and all backgrounds.

The source catalogues have different physical fluxes, transport exposures, and source-level reweighting. Each event therefore carries a physical-rate weight, and the signal or background rate at a given selection stage is obtained by summing the weights of events that pass that stage. Raw record counts quantify Monte Carlo transport support rather than physical rate. Finite-transport statistical uncertainties are propagated from the actual weight distribution, while focused-signal acceptance is treated binomially using the common focal-plane

sample. Long-duration rate propagation along the mission trajectory is described in Section 3.4; the response-convolved spectra through the active-veto and topology/aperture stages are reported with the cut-flow results.

3.3.3 Active anticoincidence gate

Active anticoincidence rejects events that produce a 511 keV candidate in the TES while also depositing energy in the surrounding active shield. Passive shielding attenuates incident radiation, whereas active BGO provides time-coincident information for event-by-event discrimination. A focused 511 keV photon reaches the TES mainly through the optical channel. Atmospheric charged particles, secondaries from hadronic interactions, and some high-energy atmospheric- γ cascades are more likely to deposit energy in both the TES and the surrounding active volumes. Coincident BGO energy can therefore identify background candidates in the science window.

The active gate uses the common time axis and $1\ \mu\text{s}$ coincidence groups defined in Section 3.3.1. Within each candidate group, TES deposits are first combined by pixel and then passed through the response defined in Section 3.3.2; the science-window decision uses the sum of the measured energies of all retained pixels. Simulated energy deposits in all BGO volumes in the same group are summed as E_{BGO} . This active-volume sum is used only for anticoincidence and does not contribute to the TES event energy.

The active system of mass model A comprises a BGO side shell, bottom cap, and top annulus. The active system of mass model B comprises the chimney side shield, front optical annulus, and rear cold-port annulus. Both mass models use the same group-level passing condition,

$$E_{\text{BGO}} < 50\ \text{keV}.$$

The common offline threshold is applied to the group-summed BGO deposit; $E_{\text{BGO}} \geq 50\ \text{keV}$ vetoes the group. Within a given mass model, the same group-level criteria are applied to the focused signal, eight-family prompt background and delayed activation.

The groupwise sums include both correlated TES–shield deposits from one incident particle and accidental overlap of otherwise independent hits whose arrival times satisfy the $1\ \mu\text{s}$ linking rule. A focused photon can therefore be rejected when an unrelated active-shield hit enters the same coincidence group. The associated loss is described by the accidental signal-survival fraction defined in Section 3.3.1 and included in the mission-time fold. Candidates that pass the active gate proceed to the Compton-topology and entrance-aperture consistency selection.

3.3.4 Compton topology and entrance-aperture consistency

Active anticoincidence removes candidates with energy deposits above the adopted threshold in the surrounding scintillators, but it cannot distinguish a focused photon from a background event whose interactions remain within the TES array. Either can leave a total energy near 511 keV in the focal plane. A focused photon must, however, arrive through the side-entry optical channel. For an event with two or more separated TES hits, the division of energy among the hits and their relative positions constrain the possible arrival directions. We therefore test whether each event is compatible with the entrance aperture. The test uses the kinematic cone construction of a Compton camera [30], but only as an aperture-consistency veto, not for imaging or source localization.

The six-layer pixelated TES focal plane provides the information needed for this test at pixel scale. For the retained hits defined in Section 3.3.2, the response model supplies a measured energy, while the event catalogue preserves the TES layer and pixel position. Hits in different pixels or layers therefore provide both an energy division and a finite spatial lever arm. The catalogue does not determine the interaction order or the position within a pixel. A single-hit event consequently carries no Compton-direction information and is retained as a calorimetric line candidate; multi-hit events proceed to the consistency test below.

For an assumed interaction order, let a candidate of total energy E_0 deposit E_1 at the first representative position $\mathbf{r}_1$ and then propagate with $E_{\text{sc}} = E_0 - E_1$ to a second position $\mathbf{r}_2$. The measured energies give the Compton scattering angle

$$\cos \theta_C = 1 - m_e c^2 \left(\frac{1}{E_{\text{sc}}} - \frac{1}{E_1 + E_{\text{sc}}} \right). \quad (9)$$

The vector $\mathbf{r}_2 - \mathbf{r}_1$ fixes the scattered-photon direction $\hat{\mathbf{u}}_{12}$, but not the incident direction. Together, $\hat{\mathbf{u}}_{12}$ and θ_C define a cone of allowed incident directions with apex $\mathbf{r}_1$ and axis $-\hat{\mathbf{u}}_{12}$. The event is aperture compatible if this cone intersects a circular analysis disk in the side-entry plane. The disk radii are 1.898 cm for mass model A and 1.900 cm for mass model B. These disks define the geometric region used by the selection; they are not the complete physical windows or collimator openings.

The remaining position uncertainty is handled explicitly. The catalogue gives an analysis position for each hit, but does not resolve the interaction point within the pixel. We represent that uncertainty by an axis-aligned box centred on the recorded position, with half-lengths 0.075, 0.075, and 0.150 cm along the analysis x , y , and z axes. Its eight vertices and centre give nine representative positions, not nine additional measurements. For an assumed order of the first two hits, the two boxes produce $9 \times 9 = 81$ position pairs and hence 81 candidate cones. Each cone is sampled at 24 uniformly spaced azimuths and extended forward to the entrance plane. The cone intersects the disk if a sampled point lies inside it or if a segment joining adjacent samples crosses it. The test is deliberately permissive: one intersection among the 81 candidate cones is enough to retain that order. Figure 6 summarizes this construction.

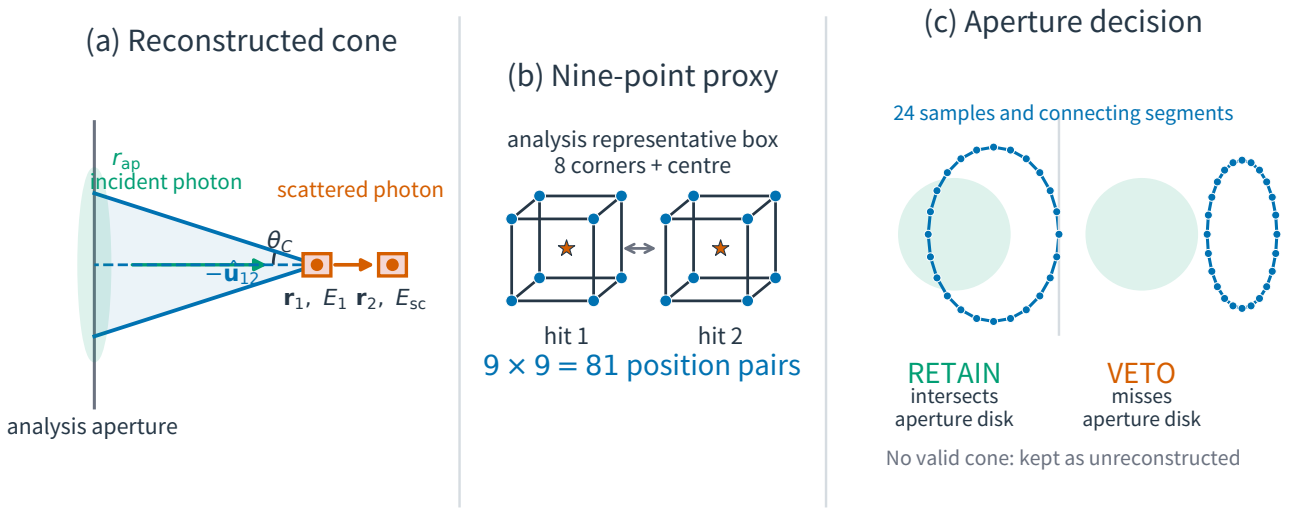


Figure 6: Schematic of the Compton-cone/aperture-consistency veto. (a) The first two TES hits define a reconstruction cone from their measured energies and representative positions; the cone is a set of allowed source directions, not a unique photon trajectory. (b) Each hit is represented by the eight vertices and centre of an axis-aligned analysis box, giving $9 \times 9 = 81$ position hypotheses for a specified order. (c) The cone trace in the entrance plane is approximated by 24 azimuthal samples and their connecting segments. An intersection with the analysis-aperture disk retains the event, whereas a valid reconstruction that misses the disk is vetoed. Events without a valid cone remain in the rate accounting as unreconstructed. The schematic is not to scale.

The interaction order is treated separately from the within-pixel position uncertainty. For a two-hit event, both orders are tested and either compatible order is sufficient. For an event with at least three hits, all orders are first enumerated using the recorded analysis positions. For a candidate sequence of n hits $\{(\mathbf{r}_j, \epsilon_j)\}_{j=1}^n$, the Compton sequence residual is

$$Q = \sum_{i=2}^{n-1} (\cos \theta_{C,i} - \hat{\mathbf{u}}_{i-1,i} \cdot \hat{\mathbf{u}}_{i,i+1})^2, \quad (10)$$

where $\theta_{C,i}$ is calculated from the energy ϵ_i deposited at hit i and the remaining energy $\sum_{k=i+1}^n \epsilon_k$; the two unit vectors describe the adjacent flight segments. Orders without a physical Compton solution do not enter the ranking. We select the order with the smallest Q ; if two orders are tied, we choose the one with the longer first lever arm. Only the first-scatter cone of this selected order receives the 81-position aperture test.

The final decision is conservative. A reconstructable multi-hit event is vetoed only when the cone or cones tested under the applicable ordering rule miss the analysis aperture. If no candidate order has a physically valid solution, the event is marked as unreconstructed and retained in the final rate rather than rejected. In this way, failure to reconstruct an event is not itself treated as evidence that the event is background.

The veto efficiency obtained with the present Compton-trajectory analysis should be regarded as a conservative lower bound. Offline analysis may achieve higher veto efficiency by optimizing the trade-off between signal retention and background rejection. Increasing the spacing between adjacent TES layers would lengthen the Compton-reconstruction lever arm of multilayer events and could further improve veto efficiency.

3.4 From the day-15 reference result to mission-time evolution

Detector transport supplies component-response templates at 38 km. The mission fold holds these responses fixed while propagating the focused signal, eight prompt families, and delayed activation over the evolving environment and inventory [17, 34].

The synthetic reference trajectory spans 0–20 d in 81 nodes separated by 0.25 d. Its altitude ranges from 36.25 to 41.25 km, its latitude from 33.75° to 34.25° N, and its longitude from 99.75° to 100.25° E. At day 15, the reference state is 38.75 km, 34° N, and 100° E. At 34° N, the Galactic center culminates at 27°, giving the minimum line-of-sight atmospheric column toward the source. Altitude changes affect both the incident particle field and the atmospheric transmission at 511 keV; latitude and longitude provide a controlled geomagnetic modulation. The trajectory provides the same 20 d on-source exposure for both mass models; source visibility, Earth occultation, and repointing are not folded.

For a specified top-of-atmosphere 511 keV line flux, the focused-signal rate away from day 15 is scaled by the atmospheric transmission,

$$S_g^{\text{dir}}(t) = S_g^{\text{dir}}(t_{15}) \frac{T_{\text{atm},511}(t)}{T_{\text{atm},511}(t_{15})} = F_{\text{TOA},511} A_{\text{eff},g} T_{\text{atm},511}(t), \quad (11)$$

where g denotes the mass model and $A_{\text{eff},g}$ includes the detector response and the single-event acceptance of the active anticoincidence and Compton topology/aperture selections. In the plane-parallel approximation, $T_{\text{atm},511}(t) = \exp[-\mu_{\text{air}} X(t) / \sin \epsilon]$. The day-15 vertical residual column is 3.461 g cm^{-2} ; with $\epsilon = 27^\circ$ and an effective dry-air mass attenuation coefficient of $0.08736 \text{ cm}^2 \text{ g}^{-1}$ [24], this gives $T_{\text{atm},511}(t_{15}) = 0.514$. The fold recalculates this transmission at every trajectory node.

Prompt background is scaled separately for each incident family a ,

$$R_{\text{prompt},g,a}(t) = R_{\text{prompt},g,a}(t_{15}) \frac{\Phi_a(t)}{\Phi_a(t_{15})} = \Phi_a(t) K_{g,a}, \quad (12)$$

where $\Phi_a(t)$ is the node flux and $K_{g,a}$ is the selected response coefficient of the transport template. The within-family energy and angular templates remain fixed; full transport is not repeated at the 81 nodes.

Delayed activation cannot be obtained by multiplying the total day-15 delayed rate by one scale because the retained nuclides have different production histories and half-lives. The production rate of source-parent k from family a is

$$P_{g,a,k}(t) = P_{g,a,k}(t_{15}) \frac{\Phi_a(t)}{\Phi_a(t_{15})} = \Phi_a(t) Y_{g,a,k}, \quad (13)$$

where $Y_{g,a,k}$ is fixed by the mass-model-specific production catalogue. The production rate is interpolated linearly between adjacent nodes and convolved exactly with radioactive decay,

$$\begin{aligned} N_{g,a,k}(t_{\ell+1}) &= N_{g,a,k}(t_\ell) e^{-\lambda_k \Delta t_\ell} \\ &\quad + \int_0^{\Delta t_\ell} P_{g,a,k}(t_\ell + u) e^{-\lambda_k(\Delta t_\ell - u)} \text{d}u, \\ A_{g,a,k}(t_\ell) &= \lambda_k N_{g,a,k}(t_\ell). \end{aligned} \quad (14)$$

The selected delayed rate is

$$R_{\text{delayed},g}(t) = \sum_a \sum_k A_{g,a,k}(t) \varepsilon_{g,a,k}. \quad (15)$$

Advancing each family–parent component preserves its half-life and selected decay response. The initial condition is $N_{g,a,k}(0) = 0$, so the calculation does not include ground pre-activation or inventory accumulated during ascent before the first trajectory node.

The total direct background before cross-event grouping is

$$B_g^{\text{dir}}(t) = \sum_a R_{\text{prompt},g,a}(t) + R_{\text{delayed},g}(t). \quad (16)$$

Accidental coincidence between physical streams is evaluated at common-time anchors on days 0, 5, 10, 15, and 20. At each anchor, Poisson arrival times are generated from the component rates and merged onto one time axis before the $1\ \mu\text{s}$ grouping, detector response, active anticoincidence, and Compton topology/aperture selections. The calculation defines a background correction and a focused-signal survival fraction,

$$\hat{\rho}_{g,r} = \frac{n_{g,r}^{\text{tl}}/T_{g,r}^{\text{rep}}}{B_g^{\text{dir}}(t_r)}, \quad \hat{\eta}_{g,r} = \frac{n_{g,r}^{\text{pass}}}{N_{g,r}^{\text{probe}}}. \quad (17)$$

$\hat{\rho}$ converts the direct background to the final common-time rate. It can be slightly larger than unity when accidental energy summation moves an event into the science window and is therefore not a survival probability. $\hat{\eta}$ is the conditional survival fraction of a weak focused event. The two factors are interpolated separately between adjacent anchors. The equivalent exposure per anchor is $2 \times 10^4\ \text{s}$ for mass model A and $2 \times 10^5\ \text{s}$ for mass model B.

Figure 7 checks the event-time layer against the uncorrected analytic fold before applying $\rho_g(t)$. For each mass model, a separate Poisson replay is generated on days 0, 5, 10, 15, and 20 and passed through the common-time grouping and event selections. The replay rates are then compared with $B_g^{\text{dir}}(t)$ from Equation (16), evaluated on all 81 nodes. The largest replay-only differences are 1.67σ for mass model A and 1.09σ for mass model B; no anchor differs from the analytic direct-rate curve by 2σ . This agreement supports use of the analytic fold as the time-dependent rate baseline within the sampled trajectory. Each replay uses component rates from Equations (12)–(15) and is not an independent transport.

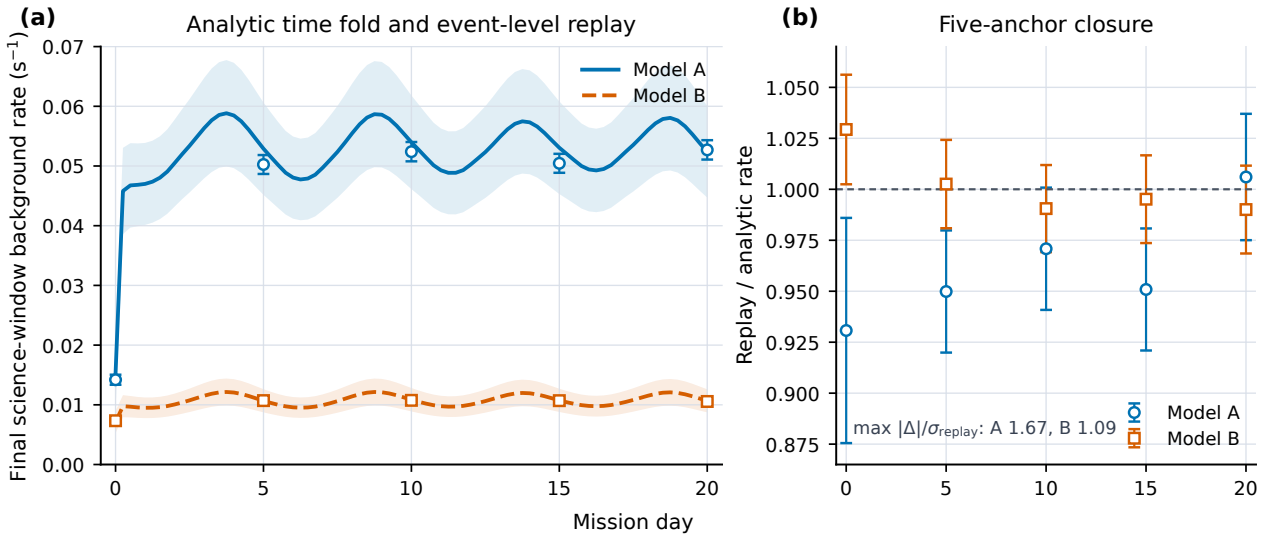


Figure 7: Event-level closure of the mission-time background fold. (a) The solid and dashed curves are the 81-node analytic direct science-window background rates obtained from the day-15 component responses, family-resolved environmental scaling, and nuclide-resolved activation evolution. Shaded bands show the finite-transport 1σ template uncertainties; because the same transport templates are reused at all nodes, these uncertainties are correlated in time. Open symbols show separate common-time Poisson replays on days 0, 5, 10, 15, and 20. Their error bars are $\sqrt{n}/T$, with $T = 2 \times 10^4\ \text{s}$ for mass model A and $2 \times 10^5\ \text{s}$ for mass model B. (b) Replay-to-analytic rate ratios with replay-only statistical errors; the horizontal line marks unity. The transport bands and replay errors are shown separately and are not combined. Replays test rate mapping, common-time grouping, and selection for supplied component rates; they are not independent transports.

The final time-dependent rates are

$$S_g(t) = S_g^{\text{dir}}(t)\eta_g(t), \quad B_g(t) = B_g^{\text{dir}}(t)\rho_g(t). \quad (18)$$

Trapezoidal integration over the 81 nodes then gives the cumulative counts at exposure T_n ,

$$\begin{aligned} N_s(T_n) &= \sum_{\ell=1}^n \frac{S_g^{\text{dir}}(t_{\ell-1})\eta_g(t_{\ell-1}) + S_g^{\text{dir}}(t_{\ell})\eta_g(t_{\ell})}{2} \Delta t_{\ell}, \\ N_b(T_n) &= \sum_{\ell=1}^n \frac{B_g^{\text{dir}}(t_{\ell-1})\rho_g(t_{\ell-1}) + B_g^{\text{dir}}(t_{\ell})\rho_g(t_{\ell})}{2} \Delta t_{\ell}. \end{aligned} \quad (19)$$

The resulting $N_s(T)$ and $N_b(T)$ are the inputs to the sensitivity definition in the next subsection. Their cumulative realization is reported in Section 4.5.

3.5 Mission significance and minimum resolvable line flux

For cumulative signal and background counts, we define the counting significance as

$$Z(T) = \frac{N_s(T)}{\sqrt{N_b(T)}}. \quad (20)$$

Under the high-count, weak-signal conditions considered here, this is the large-sample limit of the Poisson counting likelihood for $N_s \ll N_b$ [42].

The focused-signal count is linear in line flux, whereas the predicted background is independent of a weak source. The $q\sigma$ minimum resolvable top-of-atmosphere line flux at exposure T is therefore

$$F_{q\sigma}(T) = F_{\text{ref}} \frac{q}{Z(T; F_{\text{ref}})}, \quad q \in \{3, 5\}. \quad (21)$$

The time required to reach a specified significance is the first crossing of $Z(T) = q$.

The same transport events are reused at all 81 mission nodes, so their statistical errors are correlated across time. We first accumulate each event's weighted contribution over all nodes and then calculate the statistical uncertainty of the mission-integrated counts. Statistical errors from the five time anchors, effective area, and signal probes are propagated into the time curves and flux threshold. Threshold-crossing times are linearly interpolated between adjacent nodes.

4 Results

We first trace model-A backgrounds to their physical origins, then compare model B on the common mission time axis. Both models use the response and selection pipeline defined above, with geometry-specific active channels. For presentation, the prompt branch is separated into atmospheric γ rays and the seven non- γ families. Rates are sums of event weights, and the common-time replay draws Poisson arrivals from those component rates.

4.1 Baseline background and selection performance of mass model A

Mass model A defines the reference performance. Table 2 gives the day-15 direct, no-coincidence cut flow in the [510.58, 511.42) keV science window. The active-BGO gate reduces the total background from 3,086 records and 5.436 s^{-1} to 443 records and $6.071 \times 10^{-2} \text{ s}^{-1}$, removing 98.88% of the pre-veto rate. The seven non- γ prompt particle families fall from 2.912 s^{-1} to a zero central value, the atmospheric- γ component falls from 2.357 s^{-1} to $1.789 \times 10^{-2} \text{ s}^{-1}$, and delayed activation falls from 0.1675 s^{-1} to $4.282 \times 10^{-2} \text{ s}^{-1}$.

After the Compton topology/aperture selection, 400 background records remain at $5.306 \times 10^{-2} \text{ s}^{-1}$, or 87.39% of the active-veto rate. All 28,612 isolated focused-signal records in the measured window pass the active gate, and 28,006 survive the final selection. The corresponding effective area is $15.123 \pm 0.045 \text{ cm}^2$, equal to 97.88% of the measured-window signal area. Active anticoincidence supplies most of the rate reduction, while the topology selection removes a smaller residual and retains most of the focused signal.

Table 2: Mass model A science-window cut flow. Background entries are day-15 direct, no-coincidence record counts and rates; signal entries give record counts and effective area for the common 37,194-photon focal-plane sample.

Stream	Pre-veto	After active veto	Final topology/aperture
Seven non- γ prompt particle families	869 / 2.912 s^{-1}	0 / 0	0 / 0
Atmospheric γ	922 / 2.357 s^{-1}	7 / $1.789 \times 10^{-2} \text{ s}^{-1}$	6 / $1.534 \times 10^{-2} \text{ s}^{-1}$
Delayed activation	1295 / 0.1675 s^{-1}	436 / $4.282 \times 10^{-2} \text{ s}^{-1}$	394 / $3.772 \times 10^{-2} \text{ s}^{-1}$
Total background	3086 / 5.436 s^{-1}	443 / $6.071 \times 10^{-2} \text{ s}^{-1}$	400 / $5.306 \times 10^{-2} \text{ s}^{-1}$
Focused signal	28612 / 15.45 cm^2	28612 / 15.45 cm^2	28006 / 15.123 cm^2

The nonzero components of the final direct background are delayed activation and atmospheric γ rays (Table 3). Delayed activation contributes $3.772 \times 10^{-2} \text{ s}^{-1}$ (71.10%), and the atmospheric- γ component contributes $1.534 \times 10^{-2} \text{ s}^{-1}$ (28.91%). Because the event weights differ among components, record count alone does not measure statistical support; the 400-record total has $N_{\text{eff}} = 46.77$.

Table 3: Day-15 final direct background of mass model A. Standard errors and effective sample sizes are calculated from $\sum_i w_i^2$ for unequal event weights.

Component	Records	Rate / s^{-1}	Standard error / s^{-1}	N_{eff}	Total / %
Delayed activation	394	3.772×10^{-2}	4.582×10^{-3}	67.79	71.10
Atmospheric γ	6	1.534×10^{-2}	6.261×10^{-3}	6.00	28.91
Total	400	5.306×10^{-2}	7.758×10^{-3}	46.77	100.0

4.2 Background origins and spatial coupling in mass model A

The preceding subsection establishes the background level of mass model A. To identify its physical origins, we traced all final events by initiating particle family. The delayed-activation component was further decomposed by parent-production material, source-volume/parent-nuclide pair, and spatial region relative to the TES.

Proton-induced activation is the largest single contribution, with 35 records and $2.485 \times 10^{-2} \text{ s}^{-1}$, or 46.83% of the total direct background. The remaining delayed terms are $7.492 \times 10^{-3} \text{ s}^{-1}$ from α -induced activation, $3.601 \times 10^{-3} \text{ s}^{-1}$ from neutron-induced activation, $1.686 \times 10^{-3} \text{ s}^{-1}$ from γ -induced activation, $7.021 \times 10^{-5} \text{ s}^{-1}$ from e^+ -induced activation, and $2.528 \times 10^{-5} \text{ s}^{-1}$ from e^- -induced activation. Together they reproduce the final delayed rate of $3.772 \times 10^{-2} \text{ s}^{-1}$.

In the material decomposition, copper accounts for 70.87% of the final delayed rate. The three leading source-volume/nuclide pairs are ^{62}Cu produced in the TES substrate-support copper panel, the 50 mK mixing-chamber copper cold plate, and the TES-stack copper heat-sink ring. They are followed by ^{199}Po produced in the Bi passive-shield layer and ^{64}Cu produced in the same 50 mK cold plate and TES substrate-support copper panel. This ordering includes particle transport, detector response, active anticoincidence, and Compton selection.

Grouping the parent-production sites by transport component identifies the 50 mK mixing-chamber copper cold plate as the largest source, at 28.59% of the final delayed rate. The two TES substrate-support copper panels (L2 and L4) contribute 27.73%, followed by the TES-stack copper heat-sink ring at 10.18% and the near-field Bi passive-shield layer at 7.49%. Aluminium cryostat and shield components contribute 15.35% in total; the remaining dilution-refrigerator/cold-stage hardware, the tungsten detector-bay bottom plate, and the BPE plus residual BGO shielding contribute 5.26%, 3.51%, and 1.90%, respectively. An independent spatial projection assigns 32.18% of the delayed rate to the DR mixing chamber and staged cold region.

Figure 8 projects the same final selected delayed-activation events onto their parent-production components and initiating primary-particle families. The component projection resolves the cold plate, TES supports, and heat sink, while the particle-family projection identifies protons and α particles as the main drivers of this component.

Surrounding a main detector with active anticoincidence counters is an established low-background strategy in hard-X-ray and soft- γ -ray instruments; the Suzaku/HXD, for example, used thick BGO guard counters around its main detector units [37]. In the present instrument, placing conventional BGO directly at the mK cold end would add the scintillator, photosensors, mechanical supports, and cabling to the cold assembly, increasing

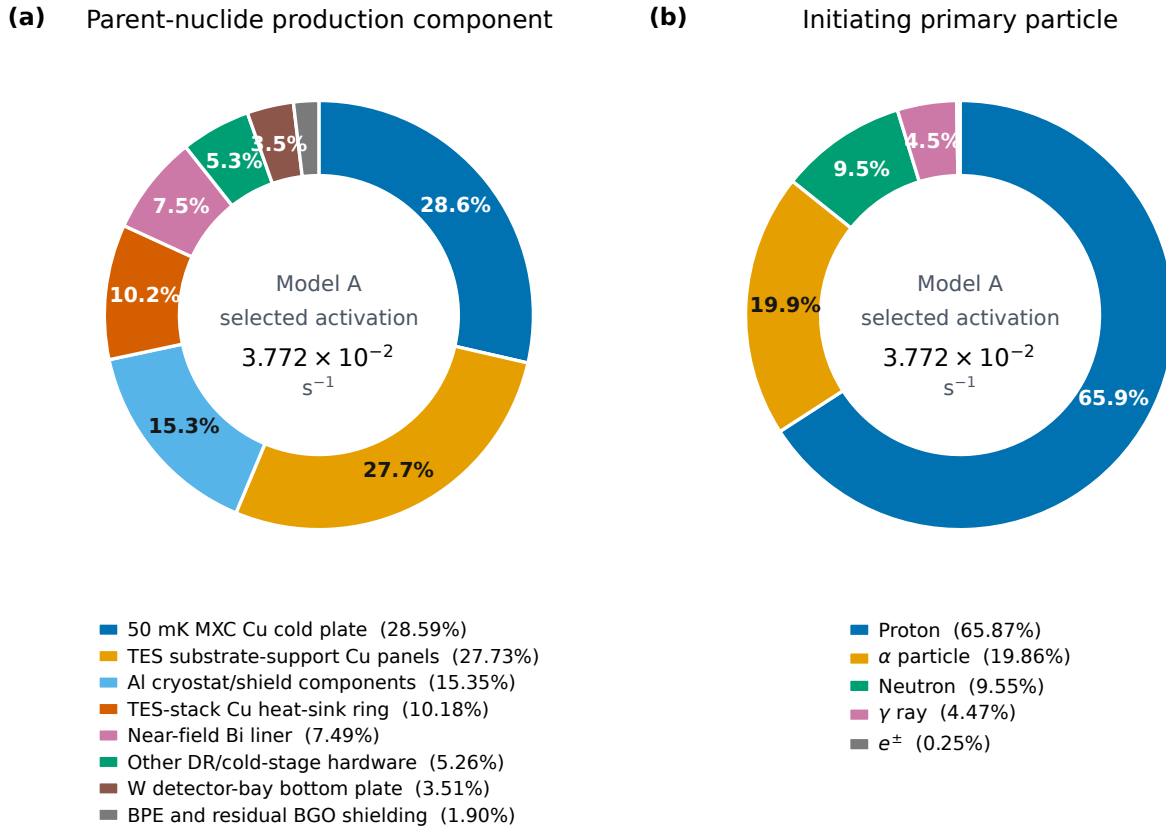


Figure 8: Origins of the final selected day-15 delayed-activation background in mass model A. (a) Component in which the parent nuclide was produced. The TES substrate-support copper category combines the L2 and L4 panels; the aluminium cryostat/shield category combines the temperature-stage shield caps, inner shield cylinder, vacuum-jacket cap, and 50 mK aluminium-can bottom cap; “other DR/cold-stage hardware” contains the remaining copper DR/cold-stage parts and the silver-sinter proxy. (b) Initiating primary-particle family, with e^+ and e^- combined as $e^\pm$. Sector areas and legend percentages are normalized to the final delayed-activation rate, $3.772 \times 10^{-2} \text{ s}^{-1}$.

both the thermal load and integration complexity. BGO light yield and pulse-decay time are also temperature dependent [39]. Mass model A therefore uses a Bi passive-shield layer near the TES while retaining the active BGO outside the ultracold assembly. At equal 511-keV attenuation, NIST XCOM gives Bi an approximately 8% lower pair-production optical depth than W for the 4.404 MeV primary traced below [24].

Several-MeV atmospheric γ rays can still pair-produce in the near-field shield and generate annihilation photons. One traced 4.404 MeV primary pair-produced in the 4.796-mm Bi layer. After the positron annihilated there, one 511.0 keV photon reached the TES, Compton-scattered, and was photoabsorbed in two pixels; the active shield recorded no energy.

This diagnosis motivates model B, which moves the TES beyond the cold-plate projection and places active BGO around the lateral focal plane.

4.3 Mass model B configuration

Mass model B places the six-layer TES focal plane laterally inside a side chimney rather than directly beneath the staged cold structures, while retaining the copper thermal-support frame of each TES layer. Heat from the layers is collected by the rear copper heat-sink assembly and conducted through a bent copper cold finger, which passes through a 1.50-cm-diameter rear cold port and connects to the 50 mK mixing-chamber cold plate. Except for this thermal-service port, the cold-stage-facing rear face of the focal plane is covered as fully as practicable by an active BGO rear annulus. Together with the chimney-side shield and front optical annulus, it tags 511-keV photons entering the rear of the TES from the cold-stage direction, including annihilation photons associated with β^+ decays in activated cold-stage copper. Estimated from the nominal geometry volumes using the same BGO density, the BGO mass decreases from approximately 298 kg in mass model A to approximately 40 kg in

mass model B, a reduction of approximately 258 kg (87%). Events retain the common $1\ \mu\text{s}$ time grouping and pass the active gate when the summed BGO deposit in the group is below 50 keV.

Figure 9 compares the two cold-end structures in the same instrument coordinates and at the same scale. The detector-neighbourhood panels emphasize the six-layer TES, layer-by-layer copper thermal supports, bent copper cold finger, rear cold port, and surrounding active BGO.

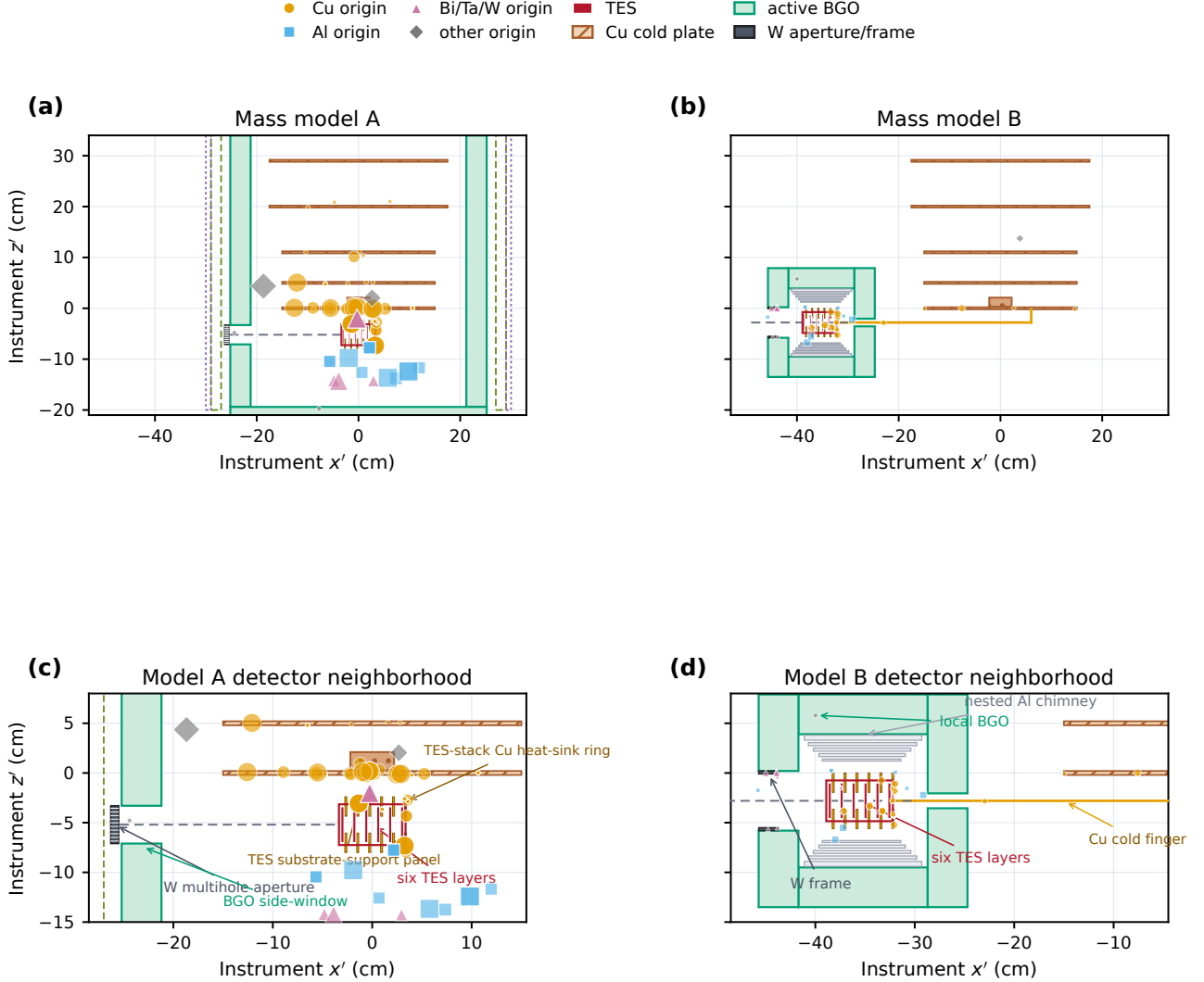


Figure 9: Dimensionally aligned x' - z' sections of the cold end and focal plane, overlaid with the parent-production positions of final delayed events. Panels (a) and (b) show mass models A and B over identical coordinate ranges and at the same scale; panels (c) and (d) show their detector neighbourhoods at a common physical scale. The drawings identify the staged copper cold plates, six-layer TES, active BGO, tungsten aperture/frame, model-B chimney and copper cold finger, and the model-A TES substrate-support copper panel, TES-stack copper heat-sink ring, and 50 mK MXC-stage copper cold plate. Each symbol marks the production position associated with a day-15 delayed event that survives the final selection; colour identifies the source material, and marker area follows the selected weighted rate on a common scale subject to a minimum display size. Sections are drawn to the transport-geometry dimensions; minor CSG apertures and service components are omitted for clarity.

4.4 Full-chain background and source redistribution in mass model B

Table 4 gives the mass-model-B day-15 direct, no-coincidence cut flow.

Active BGO reduces the total background from 7,684 records and $2.384\ \text{s}^{-1}$ to 137 records and $1.114 \times 10^{-2}\ \text{s}^{-1}$, or 0.4672% of the pre-veto rate. All 28,434 isolated focused-signal records in the measured window pass the active gate. The final Compton topology/aperture selection leaves 126 background records and retains 96.49% of the active-veto weighted rate. It retains 27,936 focused records, corresponding to $15.085 \pm 0.045\ \text{cm}^2$,

98.25% of the measured-window signal area, and 99.75% of model A's final effective area.

Table 4: Mass model B response-convolved science-window cut flow. Background entries are day-15 direct, no-coincidence record counts and rates; signal entries give record counts and effective area.

Stream	Pre-veto	After active BGO	Final topology/aperture
Seven non- γ prompt particle families	1448 / 0.8787 s^{-1}	3 / $1.811 \times 10^{-3} \text{ s}^{-1}$	3 / $1.811 \times 10^{-3} \text{ s}^{-1}$
Atmospheric γ	2987 / 1.379 s^{-1}	12 / $5.355 \times 10^{-3} \text{ s}^{-1}$	12 / $5.355 \times 10^{-3} \text{ s}^{-1}$
Delayed activation	3249 / 0.1256 s^{-1}	122 / $3.971 \times 10^{-3} \text{ s}^{-1}$	111 / $3.581 \times 10^{-3} \text{ s}^{-1}$
Total background	7684 / 2.384 s^{-1}	137 / $1.114 \times 10^{-2} \text{ s}^{-1}$	126 / $1.075 \times 10^{-2} \text{ s}^{-1}$
Focused signal	28434 / 15.35 cm^2	28434 / 15.35 cm^2	27936 / 15.085 cm^2

The final model-B direct background contains $1.811 \times 10^{-3} \text{ s}^{-1}$ from the seven non- γ prompt particle families, $5.355 \times 10^{-3} \text{ s}^{-1}$ from atmospheric γ rays, and $3.581 \times 10^{-3} \text{ s}^{-1}$ from delayed activation. Their shares are 16.85%, 49.83%, and 33.32%, respectively. Relative to mass model A, the delayed and atmospheric- γ rates fall to 9.492% and 34.92%, and the total rate is $1.075 \times 10^{-2} \text{ s}^{-1}$, or 20.26% of the model-A value, a factor of 4.94 lower. Table 5 lists the component rates and finite-transport standard errors, and Figure 12 compares the final fractional compositions of the two models using the same categories.

Table 5: Day-15 final direct-background composition and statistical uncertainty for mass model B. Standard errors and effective sample sizes are calculated from $\sum_i w_i^2$ for unequal event weights.

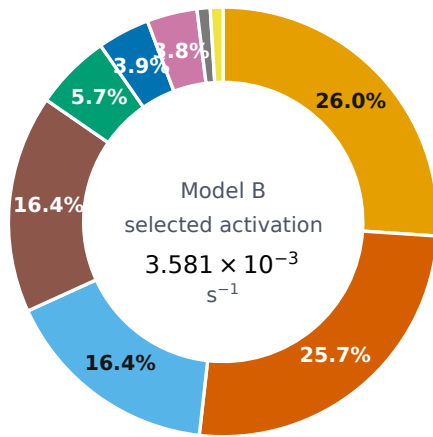
Component	Records	Rate / s^{-1}	Standard error / s^{-1}	N_{eff}	Total / %
Seven non- γ prompt particle families	3	1.811×10^{-3}	1.046×10^{-3}	3.00	16.85
Atmospheric γ	12	5.355×10^{-3}	1.556×10^{-3}	11.84	49.83
Delayed activation	111	3.581×10^{-3}	5.370×10^{-4}	44.46	33.32
Total	126	1.075×10^{-2}	1.950×10^{-3}	30.36	100.0

Copper-derived events contribute $(2.27 \pm 0.43) \times 10^{-3} \text{ s}^{-1}$, or 63.32% of the final model-B delayed rate. The spatial-region contribution from the DR/mixing chamber and staged cold plates falls from $(1.21 \pm 0.27) \times 10^{-2} \text{ s}^{-1}$ in mass model A to $(1.4 \pm 1.1) \times 10^{-4} \text{ s}^{-1}$ in mass model B. The total delayed rate changes from $(3.77 \pm 0.46) \times 10^{-2}$ to $(3.58 \pm 0.54) \times 10^{-3} \text{ s}^{-1}$.

Figure 10 resolves the 111 final delayed-activation origins in mass model B. The TES substrate-support Cu panels and TES Cu heat-sink ring contribute 26.05% and 25.73%, respectively, or 51.78% together. The W focal-plane frame and the Al chimney shells, service lid, and annuli each contribute 16.42%. The TES bottom Cu cold-plate spoke, 50 mK mixing-chamber Cu plate, and cold-link hardware contribute 5.72%, 3.93%, and 3.78%, respectively, while the TES-pixel Ta volumes and residual BGO side shield each contribute about 1%. The initiating primaries are dominated by protons (57.89%), followed by neutrons (20.81%) and α particles (18.39%); γ rays and leptons contribute 2.83% and 0.09%, respectively. Thus, the residual delayed background in mass model B is concentrated in the TES support panels and heat-sink ring, whereas the 50 mK mixing-chamber Cu plate accounts for 3.93%.

When grouped by parent nuclide, ^{62}Cu is the largest single delayed-background contributor in both mass models. Its weighted rate is $1.774 \times 10^{-2} \text{ s}^{-1}$ in mass model A and $1.238 \times 10^{-3} \text{ s}^{-1}$ in mass model B, accounting for 47.02% and 34.58% of the respective final delayed rates. Together, ^{61}Cu , ^{62}Cu , and ^{64}Cu contribute 65.50% of the final delayed rate in mass model A and 51.82% in mass model B. Figure 11 compares the leading parent-nuclide compositions.

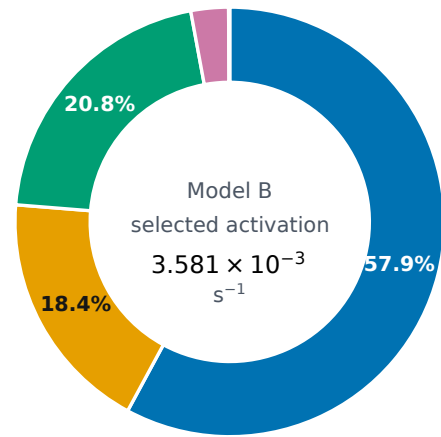
(a) Parent-nuclide production component



- TES substrate-support Cu panels (26.05%)
- TES-stack Cu heat-sink ring (25.73%)
- Al chimney shells, lid, and annuli (16.42%)
- W focal-plane frame (16.42%)
- TES bottom Cu cold-plate spoke (5.72%)
- 50 mK MXC Cu cold plate (3.93%)
- Cold-link hardware (Cu finger + SS rod) (3.78%)
- TES-pixel Ta volumes (0.99%)
- Residual BGO side shield (0.97%)

(b)

Initiating primary particle



- Proton (57.89%)
- α particle (18.39%)
- Neutron (20.81%)
- γ ray (2.83%)
- Leptons ($e^\pm, \mu^-$) (0.09%)

Figure 10: Origins of the final day-15 delayed-activation background in mass model B. (a) Component in which the parent nuclide was produced. (b) Primary particle that initiated its production. Four Cu substrate-support panels (L2/L4/L5) are grouped, as are the Al chimney/port/lid/annulus volumes and the Cu-finger/SS-rod cold link; Ta pixels and BGO remain separate. Leptons combine e^- , e^+ , and μ^- . Each of the 111 records was reweighted by primary family to the current formal rate before aggregation. Both panels use $3.581 \times 10^{-3} \text{ s}^{-1}$ as their denominator.

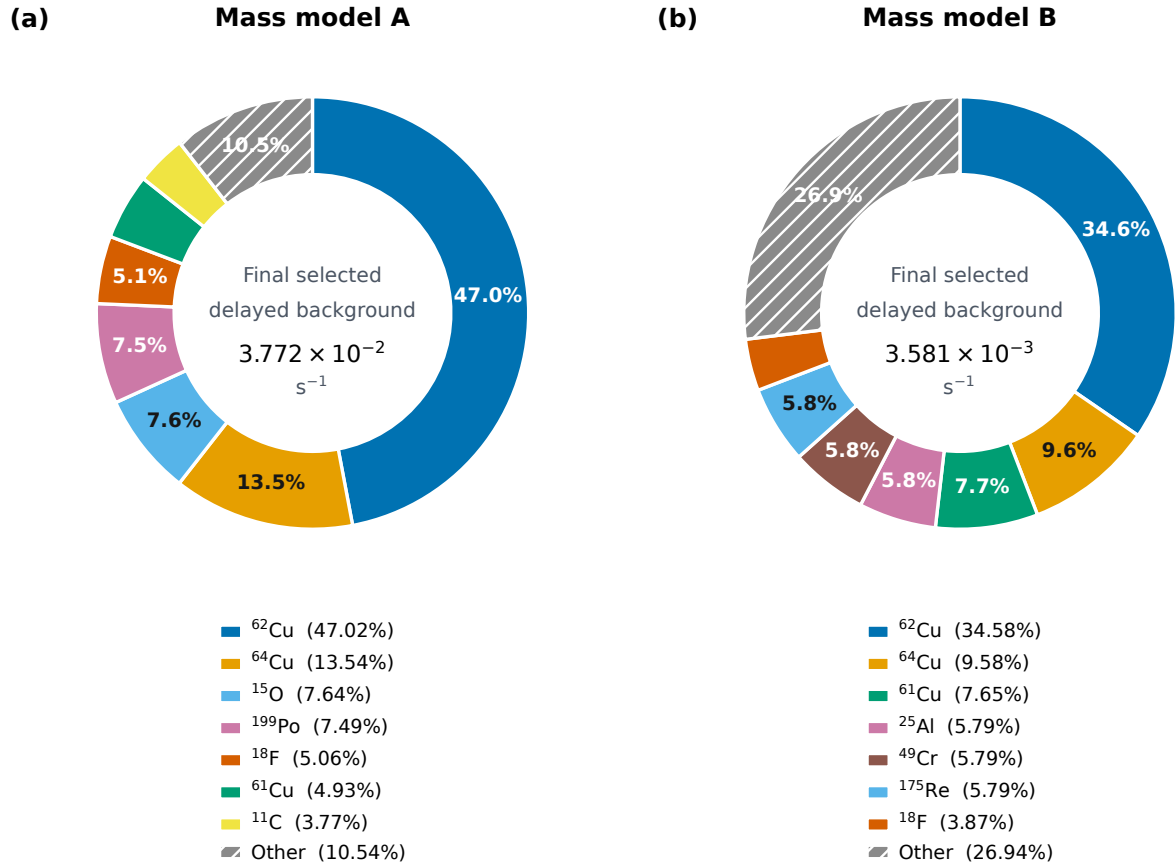


Figure 11: Parent-nuclide composition of the final selected day-15 delayed-activation background in (a) mass model A and (b) mass model B. All events have passed the TES response, the 510.58–511.42 keV science window, active anticoincidence, and the Compton topology/aperture selection. Sector area gives each parent nuclide’s fraction of the corresponding final delayed rate. The seven leading parent nuclides are shown separately in each panel and the remainder is grouped as Other. Nuclides shared by the two panels use the same colours, and the total delayed-background rate is printed at each centre. The compact model-B origin records are reweighted to the current formal rate of each initiating particle family.

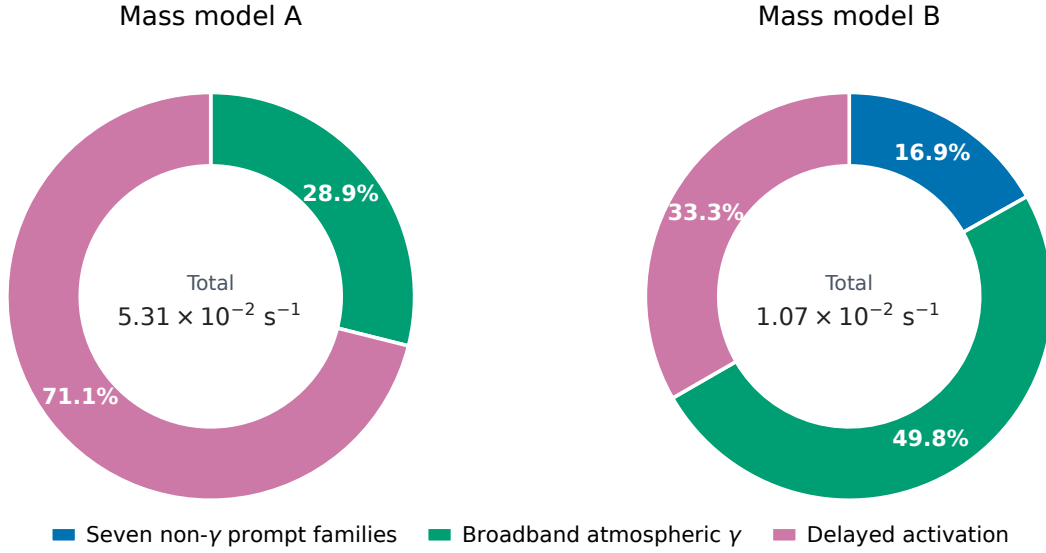


Figure 12: Composition of the final direct background in the day-15 510.58–511.42 keV science window after all selections. Sector area gives each physical component’s fraction of the corresponding model total, shown at the centre; the same component colours are used in both panels.

Figure 13 shows the day-15 spectra before active anticoincidence, after the active gate, and after the final Compton selection. The 480–550 keV band provides a selection diagnostic independent of the narrow-window integral. In mass model A, its weighted rate changes from 10.80 s^{-1} to 0.1408 s^{-1} and then 0.1213 s^{-1} ; the corresponding model-B rates are 4.922, 0.07353, and 0.06552 s^{-1} . The same ordering appears in the science-window cut flows: active anticoincidence supplies the dominant suppression, and the Compton topology/aperture selection makes a smaller correction to the residual.

The larger reduction in delayed activation shifts the dominant component from delayed activation in model A to atmospheric γ rays in model B.

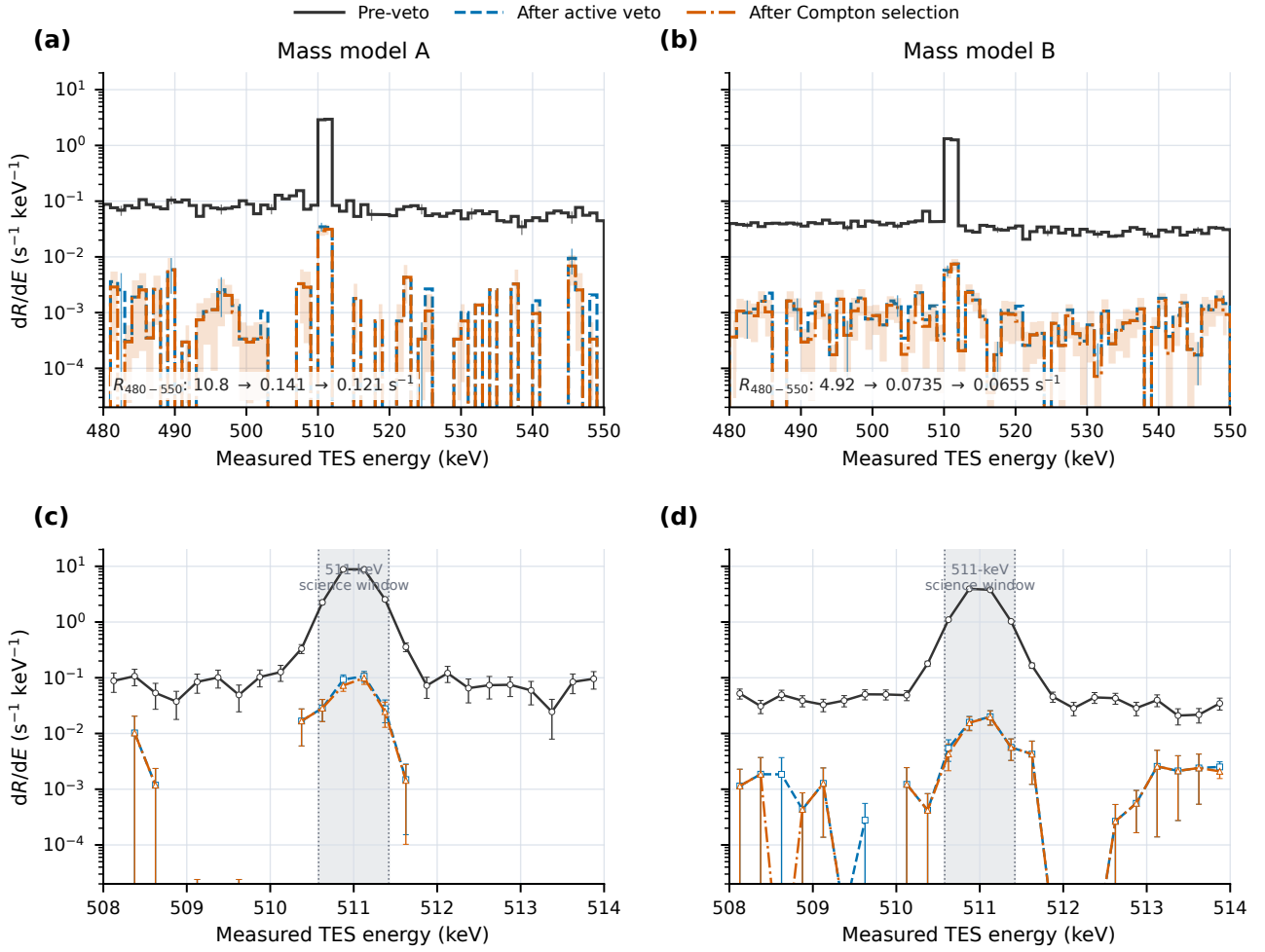


Figure 13: Day-15 detector-side differential count-rate spectra through the event-selection chain. Panels (a) and (b) show 480–550 keV for mass models A and B; four adjacent 0.25 keV analysis bins are summed into each 1 keV display bin. Panels (c) and (d) show 508–514 keV at the native 0.25 keV binning; the shaded interval and dotted boundaries mark the [510.58, 511.42) keV science window. Curves show the direct, no-coincidence background before active anticoincidence, after active anticoincidence, and after the final Compton topology/aperture selection. The translucent band in the 480–550 keV panels and the bin-wise error bars in the zoom panels show finite-transport 1σ uncertainties from $\sqrt{\sum_i w_i^2}$. Zero-weight bins are left blank, without smoothing or pseudocounts. The ordinate is the detector differential count rate after transport and TES-response convolution.

4.5 Common-time fold and cumulative counts

At the day-0, 5, 10, 15, and 20 anchors, Poisson arrivals are regenerated from the physical component rates and passed through the common time grouping, detector response, active anticoincidence, and Compton selection. The ratio of the common-time final rate to the direct rate defines $\hat{\rho}_g$, and focused-signal replay gives the accidental survival $\hat{\eta}_g$; both are interpolated to the 81 mission nodes. The model-A $\hat{\rho}_g$ values span 0.9307–1.006, and the model-B values span 0.9901–1.029 (Figure 7).

All five replay rates agree with the analytic curve within 2σ using replay-only errors, confirming common-time closure.

The focused-signal $\hat{\eta}_g$ values span 0.9681–0.9704 for mass model A and 0.9955–0.9958 for mass model B; their day-15 values are 0.9682 and 0.9955. The higher model-B survival follows from lower coincidence occupancy rather than a different signal selection. The interpolated ρ_g and η_g values are combined with the same trajectory transmission and component responses and integrated over 81 nodes at 0.25 d spacing.

Combining the day-15 conditional survival fractions with the isolated-event final effective areas and the atmospheric transmission at that node gives final signal rates of 1.805×10^{-3} and $1.851 \times 10^{-3} \text{ s}^{-1}$ for mass models A and B, respectively, at the reference flux defined in Section 1.1. Mass model B has a slightly smaller

isolated-event effective area but a smaller accidental-coincidence loss, and therefore a slightly higher final reference signal rate.

At 20 d, the expected signal/background counts are 3098/87,632 for mass model A and 3176/18,569 for mass model B. Atmospheric γ rays and the remaining background contribute 25,760/61,872 counts in model A and 9,359/9,209 in model B. The model-B cumulative background is 21.19% of the model-A value.

4.6 Mission significance and minimum resolvable line flux

Figure 14 and Table 6 summarize the mission performance. At F_{ref} , the 20 d counting significances are 10.47 for mass model A and 23.31 for mass model B. The corresponding 3σ minimum resolvable top-of-atmosphere fluxes are $(6.88 \pm 0.51) \times 10^{-5}$ and $(3.09 \pm 0.28) \times 10^{-5} \text{ ph cm}^{-2} \text{ s}^{-1}$.

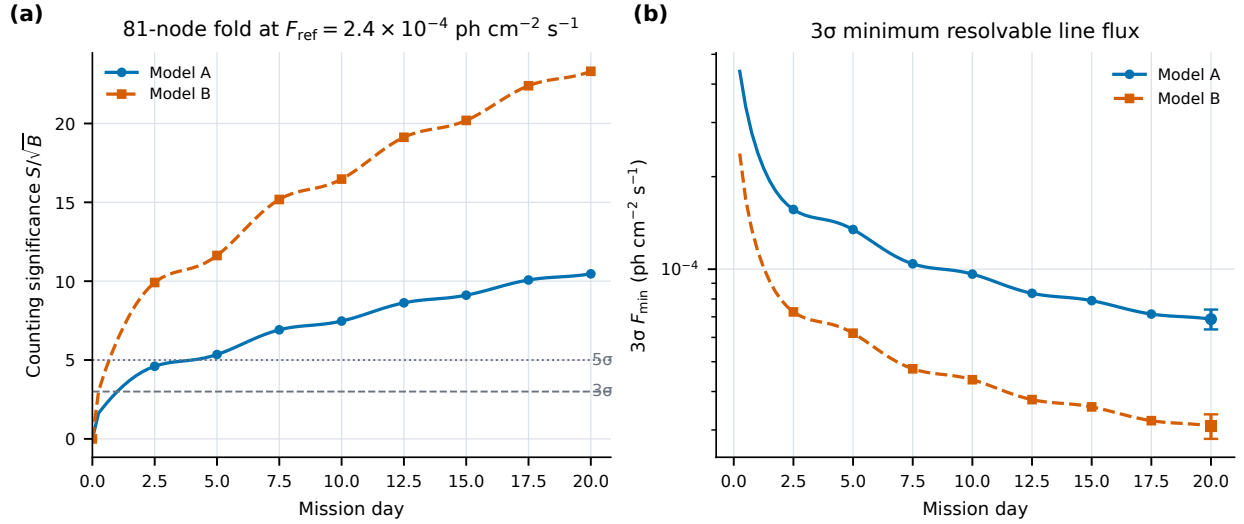


Figure 14: Conditional 81-node mission performance of mass models A and B. (a) Counting significance at the top-of-atmosphere reference flux $F_{\text{ref}} = 2.4 \times 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$; horizontal lines mark 3σ and 5σ . (b) Corresponding 3σ minimum resolvable top-of-atmosphere flux. Both designs use the same source, detector response, science window, timing, topology/aperture test, and atmospheric-transmission conventions; their entrances, local tungsten structures, and geometry-specific active channels differ.

Table 6: Conditional 20 d on-source projection for mass models A and B.

Quantity	Mass model A	Mass model B
Selected effective area / cm^2	15.123 ± 0.045	15.085 ± 0.045
Day-15 direct background / s^{-1}	5.306×10^{-2}	1.075×10^{-2}
Day-15 signal at F_{ref} / s^{-1}	1.805×10^{-3}	1.851×10^{-3}
20 d background counts	87632	18569
20 d signal counts at F_{ref}	3098	3176
$Z_{20\text{d}} = N_s / \sqrt{N_b}$	10.47	23.31
$T_{3\sigma}$ at F_{ref} / d	0.997	0.248
$T_{5\sigma}$ at F_{ref} / d	4.00	0.673
$F_{3\sigma}(20 \text{ d})$ / $\text{ph cm}^{-2} \text{ s}^{-1}$	$(6.88 \pm 0.51) \times 10^{-5}$	$(3.09 \pm 0.28) \times 10^{-5}$
$F_{5\sigma}(20 \text{ d})$ / $\text{ph cm}^{-2} \text{ s}^{-1}$	$(1.147 \pm 0.085) \times 10^{-4}$	$(5.15 \pm 0.47) \times 10^{-5}$

The flux-threshold ratio separates into cumulative-background and cumulative-signal-kernel terms. The 20 d model-B background is 21.19% of the model-A value, while its cumulative signal kernel is 1.025 times the

model-A value:

$$\frac{F_{\min,A}}{F_{\min,B}} = \sqrt{\frac{N_{b,A}}{N_{b,B}} \frac{K_B^{20d}}{K_A^{20d}}} \simeq 2.172 \times 1.025 \simeq 2.227. \quad (22)$$

Here K_g^{20d} is the cumulative signal kernel integrated over the 81 nodes. Equation (22) shows that most of the 2.227-fold threshold difference follows from the square root of the background ratio, with a smaller correction from the cumulative signal kernel.

Mass model B retains 99.75% of the focused effective area of mass model A and reduces the 20 d cumulative background and 3σ flux threshold to 21.19% and 44.90%, respectively.

5 Response screening in orbital and lunar environments

The balloon calculation establishes the model-B response and trajectory-folded sensitivity. To identify which radiation fields would limit the same detector response in other mission environments, we traced the selected events back to their primary particle types and energies and reweighted them with source spectra for LEO, the lunar surface, and Sun–Earth L2. The calculation therefore shows how a change in radiation environment acts on the established TES response and helps set priorities for detailed follow-up simulations.

5.1 Spectral reweighting

We considered four representative source fields: quiet, near-equatorial LEO at 530 km outside the SAA; an exposed lunar surface; and two Sun–Earth L2 conditions representing solar maximum and solar minimum. The LEO spectra use the components adopted in low-Earth-orbit gamma-ray background studies [35]. The lunar field combines the upward REDMoon spectrum for Apollo-17 regolith with the cosmic gamma-ray sky visible from the surface [36]. The L2 fields contain gamma rays, protons, alpha particles, electrons, and positrons, with GCR spectra appropriate to the two solar-modulation conditions [38, 40, 41].

For the prompt-gamma background, each selected event retains the energy of its primary photon. A selected delayed event is linked instead to the activation-production energy distribution for the same incident particle, parent nuclide, and production region. At those primary energies, each balloon event weight is multiplied by the ratio of the target spectrum to the balloon spectrum. The two components are then normalized to the formal day-15 rates of mass model B: $5.355 \times 10^{-3} \text{ s}^{-1}$ for prompt gamma rays and $3.581 \times 10^{-3} \text{ s}^{-1}$ for delayed activation. The principal response bands shown in Figure 15 are 0.598–29.6 MeV for selected prompt gamma rays and 1.37–48.4 GeV for proton-induced delayed background.

The seven non-gamma prompt families contribute 16.85% of the balloon background. Because the transfer response lacks an independent prompt-positron energy kernel, their aggregate rate follows the combined prompt-gamma and delayed-activation scaling; no family-resolved target-environment rates are reported.

All scenarios use the same 20 d exposure and the effective area of mass model B. The screening flux is anchored to the balloon result of $3.089 \times 10^{-5} \text{ ph cm}^{-2} \text{ s}^{-1}$, scaled with the square root of the reweighted background and corrected for the cumulative signal loss of the balloon trajectory relative to an unattenuated, continuously on-source observation. The balloon value is quoted at the top of the atmosphere, whereas the other values refer to the local instrument incident plane.

5.2 Screening results for LEO, the lunar surface, and L2

Table 7: Spectral-reweighting screen for mass model B in different radiation environments. The third column gives the combined prompt-gamma and delayed-activation rate relative to the balloon reference; the last column is the 3σ screening flux for the same 20 d exposure.

Scenario	Source-field proxy	Reweighted background/balloon	20 d 3σ screening flux / $\text{ph cm}^{-2} \text{ s}^{-1}$
Balloon reference	formal 81-node trajectory	1.000	3.089×10^{-5}
530 km LEO	near-equatorial, quiet non-SAA	0.5627	1.176×10^{-5}
Exposed lunar surface	REDMoon Apollo-17 regolith	23.02	7.525×10^{-5}
Sun–Earth L2 (2014)	solar maximum, GCR-low	5.946	3.824×10^{-5}
Sun–Earth L2 (2009)	solar minimum, GCR-high	11.92	5.414×10^{-5}

For quiet non-SAA LEO, the reweighted background is 0.5627 times the balloon reference. After removing the signal loss associated with the balloon atmosphere and observing trajectory, the 20 d screening flux is $1.176 \times$

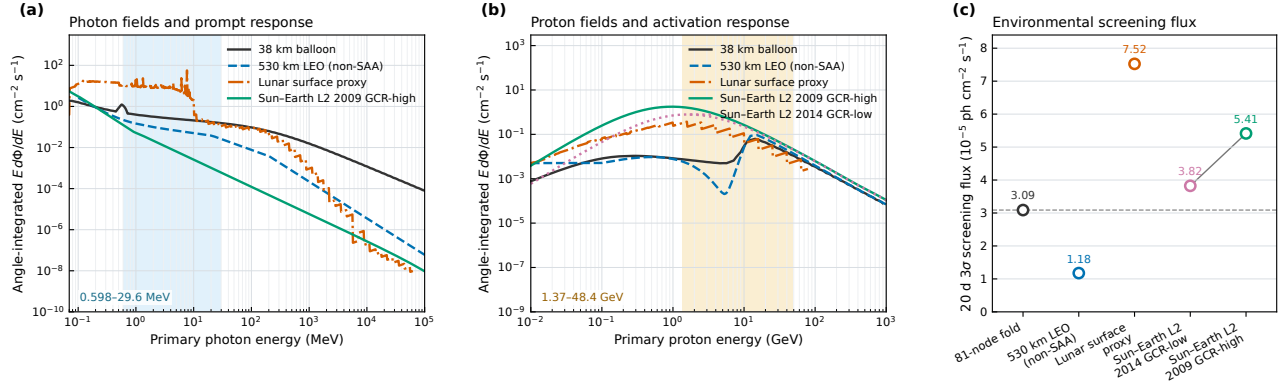


Figure 15: Environmental spectral reweighting for mass model B. (a) Gamma-ray source spectra; the shaded band is the principal primary-energy range of the selected prompt-gamma events (0.598–29.6 MeV). (b) Proton spectra; the shaded band is the principal response range of the proton-induced delayed background (1.37–48.4 GeV). (c) The 20 d, 3σ screening flux anchored to the balloon result. The balloon value is defined at the top of the atmosphere and the others at the local instrument incident plane.

$10^{-5} \text{ ph cm}^{-2} \text{ s}^{-1}$. Thus, in quiet orbital segments, the current source field produces less background in mass model B than the balloon environment. For a 550 km equatorial orbit, Cumani et al. estimated that the region occupied by trapped particles above 0.1 MeV accounts for about 18% of the orbital period; science observations are normally suspended during these SAA passages [35]. A 20 d calendar interval would therefore provide about 16.4 d of live exposure before any post-SAA cooldown. The 20 d value in Table 7 denotes live science exposure. The next LEO calculation should concentrate on activation by SAA and trapped particles and on the delayed background after an SAA passage.

For the exposed lunar surface, the reweighted background rises to 23.02 times the balloon reference and the screening flux becomes $7.525 \times 10^{-5} \text{ ph cm}^{-2} \text{ s}^{-1}$. The increase is driven mainly by upward regolith radiation in the response band of the selected gamma-ray events. A lunar application therefore requires a local gamma-ray field that couples the regolith, detector, and lander structure rather than an isolated-payload model.

At Sun–Earth L2, the result follows solar modulation. The 2014 solar-maximum/GCR-low field gives $3.824 \times 10^{-5} \text{ ph cm}^{-2} \text{ s}^{-1}$, whereas the 2009 solar-minimum/GCR-high field gives $5.414 \times 10^{-5} \text{ ph cm}^{-2} \text{ s}^{-1}$. Their ordering tracks the proton spectra and identifies GCR-proton activation as the main L2 driver. Because the present delayed response is supported by a limited set of activation templates, these values are most useful for setting the priority of a dedicated proton-activation calculation.

The reweighting therefore points to a different next step in each environment: an orbital history including SAA and trapped particles for LEO, local regolith–lander coupling for the lunar surface, and higher-statistics GCR proton activation at L2. Matched prompt, activation, and delayed transport in each target environment is the next step toward mission-level performance predictions.

6 Summary

The day-15 selected background rate of mass model A is $5.306 \times 10^{-2} \text{ s}^{-1}$, with delayed activation as the largest component. Independent spatial and material projections assign 32.18% of the delayed rate to the DR mixing chamber and staged cold region and 70.87% to copper, respectively. The leading structures are the 50 mK copper cold plate, the TES substrate-support copper panels, and the TES copper heat-sink ring. The science-window background is governed by the distance, direct lines of sight, and shield coverage between activated material and the TES.

Mass model B places the six TES layers laterally in a BGO-surrounded chimney. Its final day-15 background rate is $1.075 \times 10^{-2} \text{ s}^{-1}$, composed of 49.83% atmospheric gamma rays, 33.32% delayed activation, and 16.85% from the seven non-gamma prompt families. The active BGO reduces the direct science-window rate from 2.384 to $1.114 \times 10^{-2} \text{ s}^{-1}$, after which the topology and aperture selection leaves $1.075 \times 10^{-2} \text{ s}^{-1}$. The selected effective area of mass model B is 99.75% of that of mass model A, while its final topology and aperture selection retains 98.25% of the focused signal in its own measured-energy window. Its 20 d, top-of-atmosphere

3σ minimum resolvable line flux is $(3.09 \pm 0.28) \times 10^{-5} \text{ ph cm}^{-2} \text{ s}^{-1}$, a factor of 2.23 lower than for mass model A.

The results support two priorities: reduce coupling between the TES and activation-prone near-field structures by reducing or relocating them and blocking direct lines of sight, and maximize active-BGO coverage within the optical path. In mass model B, the TES substrate-support copper panels and copper heat-sink ring still account for 51.78% of the residual delayed background and are the direct targets of the next near-field redesign.

7 Outlook

The most immediate extension is a dedicated LEO satellite. The preliminary screening in Section 5 identifies quiet non-SAA orbital segments as a priority for a complete mission design. Compared with a finite-duration balloon flight, a LEO satellite avoids attenuation through the balloon atmospheric column and can extend live exposure from months to years. The orbit, shielding geometry, and observing schedule should be designed together: a near-equatorial, low-inclination orbit, shutdown during SAA passages, planned post-SAA cooldown, reduced activation near the TES, and wider BGO solid-angle coverage are the first parameters to study. A mission model including the spacecraft, Earth-albedo radiation, trapped-particle and SAA histories, Earth occultation, and source visibility can then determine how far long-duration operation lowers the minimum resolvable line flux.

A lunar instrument is a more exploratory, longer-term option. The simple spectral reweighting in Section 5 exposes the pressure from upward regolith gamma rays, but it does not use the additional mass and installation volume that a lunar platform may provide. If those resources are available, the TES, cryogenic system, and lander can be optimized in a coupled mass model. Directional lower-hemisphere shielding, active anticoincidence below and beside the focal plane, and shadowing by the lander could reduce direct coupling of the upward regolith field to the TES. Coupled regolith–lander–instrument transport will be needed to determine whether such a design can suppress the regolith-albedo background while benefiting from unattenuated celestial photons and long cumulative exposure.

A further detector-level study should address neutron elastic scattering in the cryogenic Si wafer and nearby focal-plane supports. Nuclear-recoil energy deposited in these volumes can generate athermal and thermal phonons; if a fraction couples to the TES, the response may appear as a non-photon pulse or as thermal cross-talk [43,44]. Coupled neutron-transport, substrate-phonon, TES thermal-network, and measured pulse-template studies, together with neutron-beam calibration, are needed to determine whether such events can be reconstructed in the 511 keV science window and whether their pulse shapes permit rejection.

Declarations

Ethics approval and consent to participate Not applicable.

References

- [1] N. Prantzos, C. Boehm, A.M. Bykov, R. Diehl, K. Ferriere, N. Guessoum, P. Jean, J. Knoedlseder, A. Marcowith, I.V. Moskalenko, A. Strong, G. Weidenspointner, The 511 keV emission from positron annihilation in the Galaxy, *Rev. Mod. Phys.* 83 (2011) 1001–1056, doi:10.1103/RevModPhys.83.1001.
- [2] P. Jean, J. Knoedlseder, V. Lonjou, M. Allain, J.-P. Roques, G.K. Skinner, B.J. Teegarden, G. Vedrenne, P. von Ballmoos, B. Cordier, P. Caraveo, R. Diehl, P. Durouchoux, P. Leleux, R. Schanne, G. Weidenspointner, Spectral analysis of the Galactic e^+e^- annihilation emission, *Astron. Astrophys.* 445 (2006) 579–589, doi:10.1051/0004-6361:20053765.
- [3] C.A. Kierans, S.E. Boggs, A. Zoglauer, A.W. Lowell, C. Sleator, J. Beechert, T.J. Brandt, P. Jean, H. Lazar, J. Roberts, T. Siegert, J.A. Tomsick, P. von Ballmoos, Detection of the 511 keV Galactic positron annihilation line with COSI, *Astrophys. J.* 895 (2020) 44, doi:10.3847/1538-4357/ab89a9.
- [4] T. Siegert, R. Diehl, G. Khachatryan, M.G.H. Krause, F. Guglielmetti, J. Greiner, A.W. Strong, X. Zhang, Gamma-ray spectroscopy of positron annihilation in the Milky Way, *Astron. Astrophys.* 586 (2016) A84, doi:10.1051/0004-6361/201527510.
- [5] T. Siegert, R. Diehl, J. Greiner, M.G.H. Krause, A.M. Beloborodov, M. Cadolle Bel, F. Guglielmetti, J. Rodriguez, A.W. Strong, X. Zhang, Positron annihilation signatures associated with the outburst of the microquasar V404 Cygni, *Nature* 531 (2016) 341–343, doi:10.1038/nature16978.
- [6] J. Knoedlseder, P. Jean, V. Lonjou, G. Weidenspointner, N. Guessoum, R. Diehl, W. Gillard, M.J. Harris, G.K. Skinner, P. von Ballmoos, G. Vedrenne, P. Caraveo, B. Cordier, V. Schoenfelder, B.J. Teegarden, The all-sky distribution of 511 keV electron-positron annihilation emission, *Astron. Astrophys.* 441 (2005) 513–532, doi:10.1051/0004-6361:20042063.
- [7] H. Yoneda, T. Siegert, S. Mittal, Imaging the positron annihilation line with 20 years of INTEGRAL/SPI observations, arXiv:2509.01066, 2025.
- [8] T. Siegert, S.E. Boggs, J.A. Tomsick, A.C. Zoglauer, C.A. Kierans, C.C. Sleator, J. Beechert, T.J. Brandt, P. Jean, H. Lazar, A.W. Lowell, J.M. Roberts, P. von Ballmoos, Imaging the 511 keV positron annihilation sky with COSI, *Astrophys. J.* 897 (2020) 45, doi:10.3847/1538-4357/ab9607.
- [9] T. Siegert, R. Diehl, G. Khachatryan, M.G.H. Krause, F. Guglielmetti, J. Greiner, A.W. Strong, X. Zhang, Constraints on positron annihilation kinematics in the inner Galaxy, *Astron. Astrophys.* 627 (2019) A126, doi:10.1051/0004-6361/201833856.
- [10] G. De Cesare, Searching for the 511 keV annihilation line from galactic compact objects with the IBIS gamma ray telescope, *Astron. Astrophys.* 531 (2011) A56, doi:10.1051/0004-6361/201116516.
- [11] N.M. Barriere, L. Natalucci, N. Abrosimov, P. von Ballmoos, P. Bastie, P. Courtois, M. Jentschel, J. Knoedlseder, J. Rousselle, P. Ubertini, Soft gamma-ray optics: new Laue lens design and performance estimates, arXiv:0910.0488, 2009.
- [12] N.M. Barriere, J.A. Tomsick, S.E. Boggs, A. Lowell, P. von Ballmoos, Developing a second generation Laue lens prototype: high reflectivity crystals and accurate assembly, arXiv:1111.6700, 2011.

- [13] G. Weidenspointner, C.B. Wunderer, N. Barriere, A. Zoglauer, P. von Ballmoos, Monte Carlo study of detector concepts for the MAX Laue Lens gamma-ray telescope, arXiv:astro-ph/0603152, 2006.
- [14] F. Shirazi, E. Gau, M.A. Hossen, D. Becker, D. Schmidt, D. Swetz, D. Bennett, D. Braun, F. Kislat, J. Gard, J. Mates, J. Weber, N.R. Cavero, S. Chun, L. Lisalda, A. West, B. Dev, F. Ferrer, R. Bose, J. Ullom, H. Krawczynski, The 511-CAM Mission: A pointed 511 keV gamma-ray telescope with a focal plane detector made of stacked transition edge sensor microcalorimeter arrays, *J. Astron. Telesc. Instrum. Syst.* 9 (2023) 024006, doi:10.1117/1.JATIS.9.2.024006.
- [15] S. Zhang, J.-K. Xia, T. Sun, et al., Transition edge sensor based detector: from X-ray to gamma-ray, arXiv:2204.00010, 2022.
- [16] K. Hu, M. Fritts, D. Becker, D. Schmidt, F. Zhou, A. Dasgupta, F. Kislat, M. Keller, S. Chun, A.G. Detoito, E. Gau, X. Jin, D. Bennett, D. Braun, J. Gard, J.A.B. Mates, J. Nobles, D. Radomski, N. Rodriguez Cavero, R. Snodgrass, D. Swetz, J. Ullom, S. Vengalil Menon, J. Weber, K. Wimalasena, H. Krawczynski, Design of a mission to measure the shape and substructure of the 511 keV gamma-ray line from the center of the Milky Way, *J. Astron. Telesc. Instrum. Syst.* 12 (2026) 025002, doi:10.1117/1.JATIS.12.2.025002.
- [17] S. Gallego, U. Oberlack, J. Lommler, C.M. Karwin, A. Zoglauer, P. Jean, P. von Ballmoos, C. Kierans, C.C. Sleator, J.A. Tomsick, S.E. Boggs, Bottom-up background simulations of the 2016 COSI balloon flight, *Astrophys. J.* 986 (2025) 116, doi:10.3847/1538-4357/add6a0.
- [18] R. Tian, J. Mao, J. Liu, H. Jin, W. Cui, Simulation of non-X-ray background for the DIffuse X-ray Explorer (DIXE) mission, arXiv:2604.13569, 2026.
- [19] R. Caputo, F. Kislat, J. Racusin, on behalf of the AMEGO Team, AMEGO: Simulations of the instrument performance, *Proc. 35th International Cosmic Ray Conference, PoS(ICRC2017)* 783 (2018), doi:10.22323/1.301.0783.
- [20] L. Gottardi, K. Nagayoshi, A review of X-ray microcalorimeters based on superconducting transition edge sensors for astrophysics and particle physics, *Applied Sciences* 11 (2021) 3793, doi:10.3390/app11093793.
- [21] D. Yan, L.M. Gades, T. Guruswamy, U.M. Patel, O. Quaranta, A. Miceli, Modelling a transition-edge sensor X-ray microcalorimeter linear array for Compton profile measurements and energy dispersive diffraction, *IEEE Trans. Appl. Supercond.* 29(5) (2019) 1–4, doi:10.1109/TASC.2019.2906255.
- [22] L. Xie, Y. Zhang, Z. Li, Z. Liu, S. Shu, J. Zhou, X. Li, H. Li, H. Gao, Y. Gu, X. Lu, Y. Zhao, C. Liu, Beyond one-thousandth energy resolution with an AlMn TES detector, arXiv:2602.11728, 2026.
- [23] E.M. Vavagiakis, S.W. Henderson, K. Zheng, H.-M. Cho, N.F. Cothard, B. Dober, S.M. Duff, P.A. Gallardo, G. Hilton, J. Hubmayr, K.D. Irwin, B.J. Koopman, D. Li, F. Nati, M.D. Niemack, C.D. Reintsema, S. Simon, J.R. Stevens, A. Suzuki, B. Westbrook, Magnetic sensitivity of AlMn TESes and shielding considerations for next-generation CMB surveys, arXiv:1710.08456, 2017.
- [24] J.H. Hubbell, S.M. Seltzer, XCOM: Photon Cross Section Database, NIST Standard Reference Database 8 (XGAM), National Institute of Standards and Technology, doi:10.18434/T48G6X.
- [25] S. Agostinelli, J. Allison, K. Amako, et al., Geant4—a simulation toolkit, *Nucl. Instrum. Methods Phys. Res. A* 506 (2003) 250–303, doi:10.1016/S0168-9002(03)01368-8.
- [26] J. Allison, K. Amako, J. Apostolakis, et al., Recent developments in Geant4, *Nucl. Instrum. Methods Phys. Res. A* 835 (2016) 186–225, doi:10.1016/j.nima.2016.06.125.
- [27] M. Sanchez del Rio, R.J. Dejus, XOP v2.4: recent developments of the x-ray optics software toolkit, *Proc. SPIE* 8141 (2011) 814115, doi:10.1117/12.893911.

- [28] F. Guan, M. Asai, D.A. Bartkoski, M. Kleckner, Z. Harel, M. Salehpour, Adding the X-ray Bragg reflection physical process in crystal to the Geant4 Monte Carlo simulation toolkit, part I: reflection from a crystal slab, *Precision Radiation Oncology* 7(1) (2023) 59–66.
- [29] R. Reiazi, D. Lee, D. Bartkoski, M. Kleckner, M. Salehpour, G4BraggReflection for accurate modeling of Bragg reflection in perfect and mosaic crystals, *J. Phys. D: Appl. Phys.* 58 (2025) 295102, doi:10.1088/1361-6463/aded1d.
- [30] A. Zoglauer, R. Andritschke, F. Schopper, MEGAlib: The Medium Energy Gamma-ray Astronomy Library, *New Astron. Rev.* 50 (2006) 629–632, doi:10.1016/j.newar.2006.06.049.
- [31] T. Sato, Analytical model for estimating terrestrial cosmic ray fluxes nearly anytime and anywhere in the world: extension of PARMA/EXPACS, *PLoS ONE* 10 (2015) e0144679, doi:10.1371/journal.pone.0144679.
- [32] T. Sato, Analytical model for estimating the zenith angle dependence of terrestrial cosmic ray fluxes, *PLoS ONE* 11 (2016) e0160390, doi:10.1371/journal.pone.0160390.
- [33] F.G. Kondev, M. Wang, W.J. Huang, S. Naimi, G. Audi, The NUBASE2020 evaluation of nuclear physics properties, *Chinese Phys. C* 45 (2021) 030001, doi:10.1088/1674-1137/abddae.
- [34] C.A. Kierans, S.E. Boggs, J.-L. Chiu, A. Lowell, C. Sleator, J.A. Tomsick, A. Zoglauer, M. Amman, H.-K. Chang, C.-H. Tseng, C.-Y. Yang, C.-H. Lin, P. Jean, P. von Ballmoos, The 2016 Super Pressure Balloon flight of the Compton Spectrometer and Imager, arXiv:1701.05558, 2017.
- [35] P. Cumani, M. Hernanz, J. Kiener, V. Tatischeff, A. Zoglauer, Background for a gamma-ray satellite on a low-Earth orbit, *Exp. Astron.* 47 (2019) 273–302, doi:10.1007/s10686-019-09624-0.
- [36] M.I. Dobynde, J. Guo, Radiation environment at the surface and subsurface of the Moon: model development and validation, *J. Geophys. Res. Planets* 126 (2021) e2021JE006930, doi:10.1029/2021JE006930.
- [37] T. Takahashi et al., Hard X-Ray Detector (HXD) on Board Suzaku, *Publ. Astron. Soc. Jpn.* 59 (2007) S35–S51, doi:10.1093/pasj/59.sp1.S35.
- [38] S. Lotti et al., Review of the particle background of the Athena X-IFU instrument, *Astrophys. J.* 909 (2021) 111, doi:10.3847/1538-4357/abd94c.
- [39] M.-A. Verdier, P.C.F. Di Stefano, P. Nadeau, C. Behan, M. Clavel, C. Dujardin, Scintillation properties of $\text{Bi}_4\text{Ge}_3\text{O}_{12}$ down to 3 K under γ rays, *Phys. Rev. B* 84 (2011) 214306, doi:10.1103/PhysRevB.84.214306.
- [40] I.G. Usoskin, K. Alanko-Huotari, G.A. Kovaltsov, K. Mursula, Heliospheric modulation of cosmic rays: monthly reconstruction for 1951–2004, *J. Geophys. Res. Space Phys.* 110 (2005) A12108, doi:10.1029/2005JA011250.
- [41] N.V. Kuznetsov, H. Popova, M.I. Panasyuk, Empirical model of long-time variations of galactic cosmic ray particle fluxes, *J. Geophys. Res. Space Phys.* 122 (2017) 1463–1472, doi:10.1002/2016JA022920.
- [42] G. Cowan, K. Cranmer, E. Gross, O. Vitells, Asymptotic formulae for likelihood-based tests of new physics, *Eur. Phys. J. C* 71 (2011) 1554, doi:10.1140/epjc/s10052-011-1554-0.
- [43] R. Agnese et al., Nuclear-recoil energy scale in CDMS II silicon dark-matter detectors, *Nucl. Instrum. Methods Phys. Res. A* 905 (2018) 71–81, doi:10.1016/j.nima.2018.07.028.
- [44] S.V. Hull et al., Characterizing thermal background events for Athena X-IFU, *IEEE Trans. Appl. Supercond.* 33 (2023) 2100606, doi:10.1109/TASC.2023.3257131.